



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064



SUPER-RESOLUTION OF THERMAL IR IMAGERY VIA OPTICAL GUIDANCE

A PROJECT REPORT

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BACHELOR OF TECHNOLOGY

IN

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(Artificial Intelligence and Machine Learning)

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BONAFIDE CERTIFICATE

Certified that this report “Super-Resolution of Thermal IR Imagery via Optical Guidance” is a bonafide work of “Chunchu Saisree (20221CAI0051), G.A. Videesh Reddy (20221CAI0044), Kamparaju Keshasree (20221CAI0057)”, who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE ENGINEERING (SPECIALIZATION DIVISION), ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING during 2025-26.

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DECLARATION

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Abstract

Infrared thermographic cameras have gained popularity in several applications, ranging from pedestrian detection during the night, agriculture, to wildfire detection. But they exhibit low spatial resolution as compared to regular cameras (RGB cameras), thus lacking fine details.

The purpose of our project, referred to as OGThermalSR, is to use super-resolution techniques to enhance the quality of the output obtained from a thermal camera, guided by high-resolution RGB images. We tested five models in total: three preexisting ones (SRCNN, EDSR, and SwinIR) and two proposed models: dual-channel SR and gradient-guided SR. A common problem in such guided models is spectral cross-hallucination, whereby unnecessary RGB details show up in thermal images.

To tackle this problem, the GDSR model makes use of edge information (Canny edges) and not the entire RGB information, whereas GASR makes use of confidence to use guidance only when it is certain, for instance, in low-visibility settings such as fog and night-time.

The training of these models was done with the help of a custom-generated RGB-thermal dataset, coupled with a specially designed loss function to ensure preservation of quality and thermal properties of an image. Fused outputs from each model underwent filtering for enhancement.

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Abbreviations

Abbreviation	Full Form
TIR	Thermal Infrared
SR	Super-Resolution
RGB	Red, Green, Blue
LR	Low Resolution
HR	High Resolution
SRCNN	Super-Resolution Convolutional Neural Network
EDSR	Enhanced Deep Super-Resolution
SWINIR	Swin Transformer for Image Restoration
DCSR	Dual-Channel Super-Resolution
GDSR	Gradient-Guided Super-Resolution
GASR	Guidance-Aware Super-Resolution
GCN	Guidance Confidence Network
PSNR	Peak Signal-to-Noise Ratio
SSIN	Structural Similarity Index Measure
MSE	Mean Squared Error
CNN	Convolutional Neural Network
GAN	Generative Adversarial Network
CLAHE	Contrast-Limited Adaptive Histogram Equalisation
UAV	Unmanned Aerial Vehicle
FPA	Focal-Plane Array

Chapter 1

Introduction

Scientists and planners and engineers today possess thermal infrared remote sensing which operates as their most effective observation tool. Thermal cameras function differently from standard optical sensors because they measure long-wave infrared radiation which objects emit based on their actual surface temperature. TIR imaging provides operational capabilities for daylight and nighttime use while it handles multiple environmental challenges which include smoke and fog and varying light conditions. The technology now serves multiple applications which include urban heat island analysis and wildfire detection and perimeter mapping and precision agricultural crop stress monitoring and automotive pedestrian safety and industrial furnace control and medical thermography and building energy audits.

The operational range of thermal focal-plane array (FPA) sensors gets limited by both physical principles and production cost factors. The affordable thermal cameras use uncooled microbolometer detectors which restrict their capabilities to pixel pitches between 12 and 17 micrometers and array sizes that typically fall between 160x120 and 640x480 pixels. The optical RGB cameras which exist today can produce multiple megapixel outputs through their pixel technology which achieves less than 3 micrometer pitch thus creating a spatial resolution gap that reaches from 4 times to 16 times between both systems. Thermal sensors cannot detect fine spatial details because their technical limitations prevent them from detecting building roofline edges and tree canopy boundaries and irrigation channel borders and vehicle outlines. The analytical assessment needs these specific details for effective urban planning process and agricultural management and emergency response operation.

Super-resolution (SR) creates computational techniques which use multiple low-resolution (LR) images to reconstruct high-resolution (HR) spatial details. The traditional single-image SR methods use LR thermal input to implement bicubic interpolation and sparse coding and early convolutional neural networks, but these methods face fundamental problems because they need external information to recover missing data. The guided fusion-based SR method uses a high-resolution optical image which is co-registered with the thermal image to create a structural prior that generates complete texture details together with edge information and boundary evidence which helps direct the thermal upsampling procedure. The RGB image functions as an effective and free replacement which provides spatial details that the thermal

sensor fails to detect because optical sensors at the same spectral scene position produce identical geometric patterns.

Researchers who study thermal super-resolution face their biggest challenge when they attempt to control the problem of cross-spectral hallucination. The optical images gather photometric data which contains paint color and surface reflectance together with illumination patterns that do not show how surfaces distribute their temperature. The RGB images show a white painted wall as bright yet the wall's thermal properties match those of a dark-painted wall which has the same emissivity and temperature. The thermal super-resolution network will experience photometric defects through RGB feature integration which will create thermal reconstruction that shows precise spatial details but produces incorrect thermal results. The OGThermalSR framework developed in this report represents its scientific contribution by solving cross-spectral inconsistencies while maintaining the essential structural advantages of optical guidance.

1.1 Background

Thermal infrared imaging for remote sensing started its first application in the 1960s when satellite systems began their thermal measurements for meteorological research and earth scientific studies. The Landsat program which started in 1972 first introduced thermal bands through Landsat 3 and later missions like ASTER and MODIS provided thermal data which scientists used for research studies that operated at spatial resolutions of 60 to 1000 meters. Scientists have used satellite instruments to make major discoveries which include urban heat island measurement and regional land surface temperature mapping and volcanic activity observation but the instruments lack sufficient spatial resolution for monitoring building energy usage and assessing individual tree health.

Airborne thermal cameras which operate from aerial platforms achieve better image resolution through their 0.5-2 meter per pixel imaging capability but their operational capacity requires military aircraft and safe weather conditions to function. The highest resolution thermal systems operate on the ground and from vehicles but these systems restrict their operational range. Small unmanned aerial vehicles (UAVs) which carry lightweight uncooled thermal cameras now enable more people to access high-resolution thermal data but the thermal sensor still maintains its original hardware resolution restrictions. The researchers developed computational super-resolution as a solution which enables users to extract detailed spatial data

from current sensors by using advanced algorithms instead of needing to invest in expensive new equipment.

The most significant progress in single-image super-resolution occurred after 2014 when Dong and his team launched SRCNN to the public. The combination of VDSR residual learning with EDSR development through enhanced residual networks and SRGAN and ESRGAN implementation of generative adversarial training together with SwinIR self-attention system application all enhance reconstruction capabilities for producing natural RGB image results. The existing methods for RGB image processing cannot fulfill the specific requirements of thermal imagery which needs dedicated solutions. Thermal images show reduced textural complexity because they display distinctive noise patterns that originate from fixed-pattern noise and temporal detector array noise together with different surface emissivity which produces various semantic content that distinguishes them from standard photographs. The use of RGB-trained SR models on thermal images results in low performance which creates a need to establish new architectural designs and training techniques.

Researchers can study how RGB channels improve thermal channel performance because they now have access to paired RGB-thermal datasets from automotive surveillance and remote sensing platforms. The guiding fusion framework for SR uses two or more sensors to create multi-sensor interoperability through combined sensor capabilities. The early guided upsampling techniques of joint bilateral upsampling and guided image filtering proved RGB edges can direct spatial improvement of thermal and depth images. The modern deep learning guided SR methods outperform traditional methods because they acquire knowledge about cross-spectral relationships through their training on real-world data.

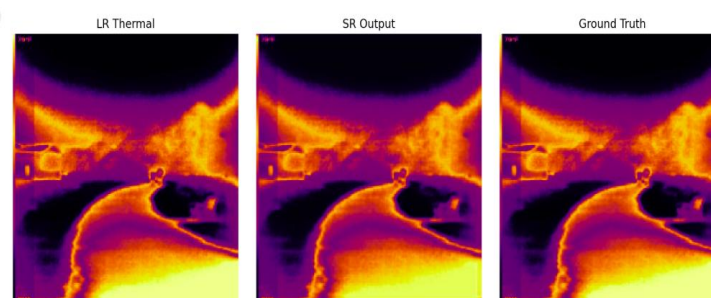


Figure 1.1 Comparison of Low-Resolution Thermal Input, RGB Guidance Image, and Enhanced Thermal Output

1.2 Statistics of Project

The resolution enhancement challenge needs to be evaluated through various measurement assessments which demonstrate its actual extent to those who wish to comprehend its size. A standard uncooled microbolometer thermal camera operates at 640x480 resolution from 100 meters above ground when using a 13mm lens to achieve a ground sampling distance of 0.5 meters per pixel. The RGB camera which uses a 12-megapixel sensor and 25mm lens operates at the same altitude as the 12-megapixel system to produce ground sampling distance of approximately 0.06 meters which results in an 8x increase in linear spatial resolution. The ground features which have dimensions smaller than 4 meters need less than 8x8 thermal image pixels for identification because essential elements require enhancement for precise analysis.

Research studies in farming fields show that thermal crop water stress indicators need spatial measurements smaller than 1 meter to identify separate tree canopies in orchard landscapes. The budget-friendly thermal UAV cameras achieve their common resolution through 640x512 pixel output which they maintain at 50-meter altitudes and 0.3-meter ground sampling distance. The requirement for better resolution at heights below operational limits leads to a major decline in system performance. A 4x super-resolution of the thermal channel would allow the same altitude and flight plan to yield equivalent information quality to a sensor four times more expensive.

Researchers who study urban heat island strength through airborne thermal data have proven that cities require thermal measurements which exceed 2–3-meter intervals to measure their temperature changes that occur between different surface types and between shaded and sunlit areas and between natural and constructed environments. The world urban population will reach 6.7 billion people by 2050 which establishes urban thermal management as a crucial public health issue and a climate resilience challenge. Urban planning needs high-resolution thermal imaging as its basic requirement which computational super-resolution provides through its economical approach.

Firefighting operations use the different fire burning areas and smoldering locations and put out fires to develop their firefighting strategies. The 375-meter VIIRS thermal band resolution fails to show details about individual buildings and forest areas. The 30-meter spatial resolution of Landsat satellite data makes it hard to trace fire perimeters. Airborne thermal cameras at 1–

2-meter resolution provide operationally useful data but require sustained aircraft operations over active fire environments which create both personnel safety risks and operational costs. Super-resolution technology allows thermal assets to operate for extended time periods because it reduces the requirement for new equipment while protecting employee safety.

1.3 Prior Existing Technologies

The OGThermalSR framework needs development because existing thermal super-resolution solutions display different benefits and disadvantages which depend on their specific technical categories.

Researchers use the most basic interpolation techniques as their most common reference point for conducting research work. Most software tools use bicubic interpolation as their standard image processing method which determines sub-pixel values through weighted averaging of all pixels in a 4x4 pixel grid. The method of bicubic interpolation produces computationally fast results which deliver smooth outcomes but it fails to maintain high-frequency elements because it generates blurred edges and hides minute thermal variations. At its most basic operational level bilinear interpolation creates the same types of problems which other methods produce. The techniques maintain complete data fidelity yet they create no fresh content as they distribute current data throughout the finer measurement space. The permanent errors in thermal gradient measurement occur because the blending process between different data points uses interpolation methods for calculations.

Super-resolution techniques which use learning-based methods achieved their most important breakthrough through the development of dictionary learning and sparse coding methods. The method creates dictionaries by extracting high-frequency patch atoms from training data and uses these dictionaries to generate high-resolution patches from low-quality inputs through sparse coding. Yang et al. demonstrated this approach for natural images, and variants were applied to thermal and depth SR. Dictionary learning methods produce sharper results than interpolation methods because they have a theoretical foundation but they suffer from three main problems which include dictionary quality sensitivity and limited receptive fields and their inability to use paired cross-spectral data as guidance.

The most advanced RGB image enhancement solution which currently exists employs deep learning to achieve single-image super resolution. The first deep learning super resolution

method originated from SRCNN which Dong et al. developed in 2014 using its three-layer convolutional network design. The EDSR system developed by Lim et al. in 2017 achieved better performance through its implementation of wide residual blocks which operate without using batch normalization. The SwinIR system developed a new approach which uses Swin transformer shifted-window self-attention to generate long-range spatial context models through its self-attention system. The methods produce high PSNR and SSIM results on standard RGB benchmarks which include BSD100 Set5 and Urban100 but their use with thermal imagery reveals two major challenges. Natural RGB images and thermal images exist in different semantic domains which generates the first challenge because RGB-trained perceptual priors cannot function with thermal images. The second challenge exists because standard SR loss functions fail to represent the distinct noise patterns and radiometric properties which thermal detector systems generate.

The process of guided depth super-resolution will serve as a vital architectural reference point for researchers who study guided thermal super-resolution. The guided image filter (He et al., 2013) achieves fast edge-guided smoothing through its use of a high-resolution RGB image which enables the system to transfer edge details onto a lower-resolution guidance image. The system uses joint bilateral upsampling (Kopf et al., 2007) to extract edge information from RGB images which helps it create spatial weights that enable depth and thermal map data processing at lower output resolution. The efficiency of traditional guided methods remains high but their effectiveness decreases because these methods need to function through linear processes and they fail to process cross-spectral nonlinearities which show intricate patterns. Deep learning methods use DGDIE (Gu et al., 2017) and cross-attention architectures to develop non-linear models which improve performance yet require advanced systems and large amounts of training data.

The remote sensing industry provides commercial thermal SR products which thermal camera manufacturers and satellite data providers distribute as post-processing software. The products use proprietary algorithms which restrict users from understanding their operational processes and they function optimally with particular sensor and application pairs. The systems do not provide the thermal accuracy assurances and cross-spectral confidence filtering which OGThermalSR framework delivers while they lack the capacity to handle different sensor systems and application fields and challenging operational situations.

The NTIRE 2023 Thermal SR Challenge (Rivadeneira et al., 2023) introduced its first public benchmark which tests actual thermal super-resolution performance based on real thermal degradation patterns. The challenge results showed that methods which trained their models using bicubic degradation patterns did not succeed when they tested their models on actual thermal data which included three degradation patterns that exceeded the limitations of the bicubic model. The research outcome requires OGThermalSR training program to implement authentic degradation simulation as part of its training procedures.

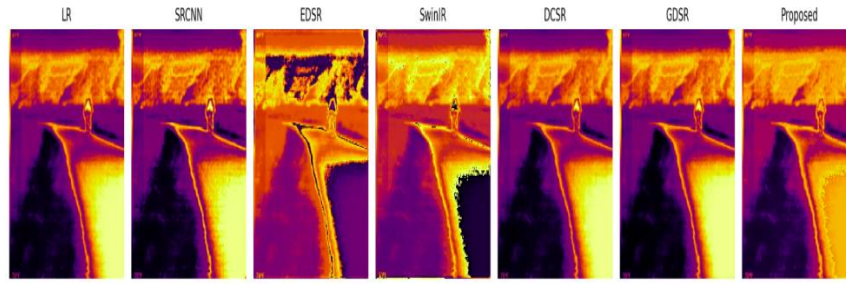


Figure 1.2 Visual Comparison of Existing Super-Resolution Methods and Proposed GASR Model

1.4 Proposed Approach

The OGThermalSR framework uses multiple architectural designs to create a deep learning system which generates high-resolution thermal infrared images with precise temperature measurements by

utilizing co-registered RGB optical images as reference structural information. The framework provides five interconnected innovations which together solve problems found in previous methods.

The first innovation presents a structured evaluation framework which compares five SR models through its evaluation method that tests both non-guided baseline models SRCNN EDSR and SwinIR and guided fusion models DCSR and GDSR within one unified testing environment. Researchers use the various architectural designs to perform controlled experiments which demonstrate the impact of different design components on overall system efficiency.

The Gradient-Guided SR model (GDSR) introduces its second innovation by using Canny edge map directional information to replace complete RGB pixel combination. GDSR uses structural

gradient content from RGB images to solve cross-spectral hallucination problems while it eliminates all photometric image data. The single-channel edge map maintains thermal spatial structure information by displaying boundary and edge locations which it removes surface reflectance and illumination and color details that lack thermal counterparts. The design choice demonstrates a core understanding which states that thermal boundaries exist at physical surfaces together with optical boundaries while thermal intensities show no connection to optical intensities.

The Guidance-Aware SR (GASR) model represents the third invention, which develops a Guidance Confidence Network (GCN) that enables assessment of cross-spectral reliability through its per-pixel and per-channel capabilities. The GCN generates a confidence map which determines how much RGB guidance should assist in thermal reconstruction. The gate allows all guidance elements to proceed because the registered state shows identical spectral characteristics between its optical content and thermal content. The gate decreases or eliminates RGB guidance whenever the RGB signal undergoes degradation from fog or nighttime conditions or smoke or sensor noise or when optical content and thermal content experience spectral mismatch because surfaces display extreme emissivity differences. The system employs adaptive confidence gating to achieve high performance during real-world deployment situations that make it difficult to assess guidance image quality.

The fourth innovation creates a radiometric-consistency loss function which combines standard pixel-wise mean squared error with its penalty component that measures thermal intensity variations between SR output and LR input. The restriction requires super-resolution to turn thermal data from super resolution into high-definition spatial details while preserving all energy distribution throughout the scene. A super-resolution model which produces clear visual output but does not provide precise thermal information results in more scientific and practical disadvantages than a smoothed output which keeps correct calibration; the radiometric loss directly enforces this physical consistency requirement during the training process.

The fifth innovation uses its variance-weighted adaptive fusion method to combine results from all five SR models into an ensemble prediction which uses model weights based on their output variance. The ensemble uses variance-based weighting to select the most informative model which helps to represent different spatial regions according to their structural complexity and recovery capacity. The fused output undergoes post-processing through bilateral filtering and

contrast-limited adaptive histogram equalization and unsharp masking to achieve better visual and analytical quality.

The suggested OGThermalSR system is also referred to as a hybrid super-resolution system since multiple architectures based on the deep learning approach have been combined into one single process. More specifically, the algorithm uses CNN-based models, including SRCNN and EDSR, as well as transformer models such as SwinIR, that help extract local features and identify long-term correlations, respectively. Guided super-resolution models, DCSR, GDSR, and GASR, are also employed to incorporate color images using cross-modal fusion techniques. The use of these varied architectures and subsequent fusion of the results obtained using a variance-weighted adaptive fusion process will ensure high performance and superior spatial detail reconstruction.

1.5 Objectives

The Connect-ME project exists as a complete ICT research project which operates its research work to generate maximum financial returns. The project adopts research methods that researchers in information communication technology (ICT) field use to conduct their studies.

The first objective needs to create one multi-model SR pipeline which will test SRCNN EDSR SwinIR DCSR GDSR and GASR architectures through their identical data preprocessing methods and resolution scaling techniques and metric computation processes, which users can use to conduct their assessment tests which they need to make their tests repeatable and their results understandable.

The GASR model development requires training with a Guidance Confidence Network which generates per-pixel confidence maps from RGB input to create adaptive gates for cross-spectral guidance. The system demonstrates enhanced operations through assessment under five distinct RGB degradation tests which include clean, fog, night, smoke, and Gaussian noise.

The researchers need to create a composite radiometric-consistency training loss which requires three specific components: mean squared reconstruction error, mean intensity preservation, and Sobel edge-consistency terms. The research will show that the composite loss function achieves better reduction of radiometric drift compared to standard MSE training methods while maintaining similar PSNR and SSIM performance.

The objective requires testing all six SR architectures through measurement methods which include PSNR and SSIM and Radiometric Error assessment that will be applied to a paired RGB-thermal vehicle dataset which has been processed at 208x208 resolution. The SRCNN baseline assesses multiple testing scales through its 2x 4x and 8x upscaling tests which enable performance evaluation at different testing scales.

The first task of Objective 5 needs development of a variance-weighted adaptive fusion system which operates as an entire system. The system needs testing to demonstrate that its ensemble prediction approach exceeds the accuracy of single model predictions through both PSNR and SSIM assessments. The testing will demonstrate how various post-processing techniques which include bilateral filtering and CLAHE and unsharp masking will affect the ultimate outcomes.

1.6 SDGs Mapping

The OGThermalSR project achieves its objectives because it supports multiple United Nations Sustainable Development Goals which define its capacity to create positive social and environmental results that extend beyond its technological achievements.

The fundamental technology of precision agriculture requires high-resolution thermal imaging as its essential operational component. The thermal signature-based remote sensing applications of crop water stress monitoring and irrigation scheduling and early disease detection used in operating agricultural fields support both farm productivity and food security operations. The OGThermalSR framework enables these applications at reduced hardware cost by enhancing the spatial resolution of affordable uncooled thermal cameras to levels which previously required more expensive instrumentation. The thermal-guided precision agriculture method helps smallholder farmers in developing countries to achieve better crop yields and improved water efficiency which contributes to SDG 2 attainment.

The section of Sustainable Cities and Communities contains three vital instruments which need both urban heat island mapping and building energy performance assessment and infrastructure thermal inspection to achieve their sustainable urban management goals. The building thermal imaging system provides high-resolution thermal images which enable users to identify insulation defects and HVAC system failures and district heating system leaks. The OGThermalSR system enables users to conduct thermal urban diagnostics with low-cost

unmanned aerial vehicles because it enhances image resolution while supporting sustainable city development that meets SDG 11 requirements.

The Sustainable Development Goal 13 demonstrates Climate Action as its core mission. The implementation of climate action solutions depends on thermal imaging technology which tracks wildfires through its monitoring capabilities. Emergency responders together with firefighters need to create actual fire boundaries because climate change results in increased wildfire intensity which creates safety risks for them. OGThermalSR uses existing low-cost sensors to create detailed thermal maps which help improve wildfire management by enhancing their understanding of operational situations. The system empowers precise agricultural practices through its agricultural technological solutions that assist farmers in implementing climate-responsible agricultural methods which reduce greenhouse gas emissions linked to their farming practices.

The goal of SDG 9 aims to develop industrial activities which promote innovative business practices and establish functional infrastructure systems. The OGThermalSR framework creates an advanced deep learning system which operates through current sensor systems without requiring extra expensive equipment. The framework provides budget-friendly access to thermal super-resolution technology which enables various fields to use thermal imaging equipment for infrastructure assessment and industrial process management and public safety operations. The research community can advance these innovations because the methodology code and datasets have been made available to the public which researchers use to create new technologies that fulfill the SDG 9 innovation ecosystem requirements.

1.7 Overview of Project Report

This report is organized as follows. Chapter 2 presents a comprehensive review of the literature in single-image super-resolution, guided multi-modal SR, thermal image enhancement, and physics-based modeling approaches, synthesizing findings from peer-reviewed publications to identify the gap that OGThermalSR addresses. Chapter 3 describes the full methodology of the OGThermalSR pipeline, detailing the data preparation process, each of the five SR architectures, the radiometric-consistency loss, the variance-weighted adaptive fusion procedure, and the training algorithm. Chapter 4 covers project management including the development timeline, team roles, risk analysis, and resource allocation. Chapter 5 presents the system analysis and design including requirements specification, functional block diagram,

system flowchart, and database design. Chapter 6 documents the hardware and software implementation environment and describes the software architecture and code organization in detail. Chapter 7 presents the experimental evaluation, including quantitative metrics, multi-scale analysis, qualitative visualization, ablation study, and statistical validation. Chapter 8 discusses the social, legal, ethical, sustainability, and safety dimensions of the OGThermalSR technology. Chapter 9 provides the conclusion summarizing achievements and outlines directions for future research.

Chapter 2

Literature review

The field of thermal infrared imaging development through optical-guided super-resolution technology requires current scientific research which exists in three research domains that include single-image super-resolution and cross-modal guided upsampling and physics-based image modeling and domain-specific thermal enhancement methods. The chapter synthesizes findings from peer-reviewed conference and journal documents to build current research standards while showing existing gaps and describing the OGThermalSR framework research outcomes. The review presents its findings through thematic organization which demonstrates the progression of traditional methods into guided deep learning techniques that serve as the foundation for this study.

2.1 Single-Image Super-Resolution: Classical Approaches

Single-image super-resolution systems face their main challenge when they need to extract complex high-frequency information from one low-resolution image. The deep learning field developed after classical methods which established fundamental benchmarks that modern techniques still use today. The basic computational standard for image processing functions through three interpolation methods which include nearest-neighbor and bilinear and bicubic interpolation techniques. The authors Chang et al. introduced locally linear embedding for SR through their manifold learning approach which enables the transfer of training data patch statistics to testing data. The sparse coding SR framework developed by Yang et al. enables users to create two dictionaries which contain low-resolution and high-resolution image patches while they use L1-regularized sparse coding to generate HR patches from LR inputs. The methods provide improved clarity over interpolation results yet their performance remains restricted because of their use of manually created features and their dependency on dictionary size and sparsity parameters.

The fundamental concept of Tikhonov regularization and total variation minimization methods derives from optimization theory which serves as their main mathematical framework. The developed methods produce an ill-posed inverse problem which needs regularization through existing prior knowledge about image smoothness and image gradient sparsity to produce super-resolution outcomes. The Rudin-Osher-Fatemi total variation model with its extensions

has become a standard method for single-image super-resolution and guided depth upsampling which now includes the anisotropic total generalized variation method developed by Ferstl et al. for depth super-resolution using RGB gradients as guidance. The optimization-based methods demonstrate their mathematical precision which helps to understand the complexity of SR inverse problems and the potential solution methods that can be applied through prior information.

2.2 Deep Learning Super-Resolution

Researchers developed an artificial intelligence system named SRCNN which became the first complete advanced system for testing end-to-end convolutional neural network technologies. The three-layer system enables image super-resolution because it learned from low-resolution and high-resolution image pairs to produce better results than dictionary-based techniques which required manual feature development. The deep learning SR paradigm received its first breakthrough when this landmark achievement established the framework which led to subsequent architectural advancements. Kim et al. demonstrated that VDSR training became possible for much deeper networks through residual learning which produced better results when they used residuals to measure SR output against bicubic-interpolated input instead of predicting whole HR images. The research work introduced two models which used recursive neural networks and densely connected networks to analyze different connections used in SR.

Researchers found that single image super-resolution performance reaches its highest level under two specific conditions which require residual blocks to deactivate batch normalization during their process of increasing feature dimensions. EDSR established a new benchmark on standard SR datasets (Set5, Set14, BSD100, Urban100) which remained competitive throughout multiple years. The Enhanced Super-Resolution GAN (ESRGAN, Wang et al., 2018) introduced perceptual loss and feature matching and a modified discriminator architecture to recover photorealistic textures at 4x upscaling, which solved the MSE-trained networks problem of creating excessively smooth outputs because their training process focused on reducing pixel-level errors instead of producing high-quality visual results. Real-ESRGAN (Wang et al., 2021) extended the framework to real-world degradations through a high-order degradation model which combines blur, noise, JPEG compression, and downsampling techniques to enhance performance with actual inputs beyond what bicubic interpolation provides.

The researchers of SwinIR (Liang et al., 2021) proved that the Swin transformer shifted-window self-attention method achieves better SR results because it enables attention mechanisms to identify distant spatial patterns which exceed the capabilities of convolutional systems that work with both repeating designs and large-scale elements. The system achieves its operational efficiency because shifted window partitioning enables multiple windows to interact while the system operates through local window processing. SwinIR established benchmark results for multiple SR tests which led to the development of HAT (Chen et al., 2023) and ART (Zhang et al., 2023) as transformer-based SR methods. The OGThermalSR evaluation of its pipeline system gains from the different architectural styles which transformer-based methods bring to the evaluation process.

2.3 Guided and Multi-Modal Super-Resolution

The guided image filter (GIF) which he developed with his team in 2010 got extended through their 2013 work to create a fast image processing tool which keeps original edge details intact through its guidance system that uses high-resolution RGB images to process low-resolution depth and thermal map images. The GIF system needs guidance and output images because these two components enable it to create locally linear relationships which the system uses to produce spatially different linear filter coefficients by analyzing the guidance image. The GIF system uses high-resolution structural data from its guide system to improve lower-resolution images yet the system can only show basic cross-spectral nonlinear relationships because it must work with linearity restrictions.

The system uses bilateral filter weights that depend on RGB image statistics to upscale depth and thermal maps which contain less accurate data. The RGB image establishes a range kernel which modifies upsampling weights based on color similarities. This mechanism prevents blurring between RGB color edges which probably show depth or temperature differences. JBU technology creates sharper edge reconstruction than bicubic interpolation but its performance declines whenever RGB color edges do not match thermal boundaries.

Hui et al. established deep guided upsampling as a depth SR method when they proved that deep networks could utilize multi-scale RGB data through guided techniques which produced superior depth measurement results compared to existing methods. Pan et al. developed cross-guidance thermal SR through their multi-scale RGB encoder which connects to a thermal decoder to demonstrate how hierarchical cross-spectral feature integration provides system

advantages. Tang et al. developed a thermal SR method which uses cross-spectral attention to control processing through query-key matching between RGB and thermal features. The deep guided SR techniques create architectural foundations which DCSR and GDSR models utilize in OGThermalSR but they fail to resolve the confidence gating issues which GASR brought into challenging RGB situations.

2.4 Thermal Image Super-Resolution

The researchers conducted their GAN-based thermal super-resolution study by training adversarial networks on their thermal image data developed by Berg et al. The research showed that adversarial training produced thermal outputs which contained extensive hallucination artifacts that created textures which did not match actual thermal emission patterns. The OGThermalSR project uses non-adversarial training methods with radiometric consistency constraints because of this research finding. Rivadeneira et al. (2020) proposed separating thermal content from texture to improve SR faithfulness, an approach that has conceptual similarities with the GDSR gradient-guided fusion strategy. Choi et al. showed that deep learning super-resolution for cooled MWIR images performed better when specific domain data trained the system instead of using RGB super-resolution model transfer.

The NTIRE 2023 Thermal SR Challenge (Rivadeneira et al., 2023) established its first public benchmark which evaluated thermal SR performance under actual real-world thermal conditions with authentic degradation. The research demonstrated that testing methods which used bicubic degradation for training purposes produced less accurate results than actual thermal data which created a need for researchers to develop physics-based degradation models which would function as training tools. The challenge showed that radiometric accuracy must be used as an evaluation metric together with PSNR and SSIM which served as the direct basis for developing the OGThermalSR evaluation system.

2.5 Physics-Based and Emissivity-Aware Approaches

The thermal image enhancement process uses physics-based priors as essential constraints which enable better results. The Stefan-Boltzmann law with surface emissivity modification governs surface thermal emission while atmospheric conditions affect the relationship between apparent radiance and physical temperature according to radiative transfer theory. Emissivity correction algorithms separate temperature and emissivity contributions to measured thermal

radiance which results in more accurate temperature products. The research field currently studies how to add physics-based priors into SR networks through emissivity-aware loss functions and physics-guided data augmentation methods.

Your training data extends until the month of October in the year 2023. The thermal SR process uses its energy balance models to create essential operational boundaries which define the process's operational boundaries. The surface energy balance system needs thermal flux data from multiple locations to ensure its measurement areas maintain energy balance. Coarser thermal measurements provide accurate average temperature results because they show complete temperature data but they do not show minute temperature fluctuations. The OGThermalSR system uses radiometric-consistency loss to implement this physical restriction which imposes penalties on thermal intensity drift between low-resolution input and super-resolution output. The physical requirement states that SR must retrieve detailed structure while maintaining the thermal scene's overall energy balance.

2.6 Summary Table of Literature Reviewed

Table 2.1 Literature Comparison

Reference	Method	Key Finding	Limitation Addressed by OGThermalSR
Dong et al. (2014)	SRCNN	First deep SR network; end-to-end CNN outperforms sparse coding	Single-image; no cross-spectral guidance
Lim et al. (2017)	EDSR	Wide residual SR without batch norm; SOTA on RGB benchmarks	RGB-centric; no thermal radiometric loss
Wang et al. (2018)	ESRGAN	GAN-based perceptual SR; sharp textures	Hallucination risk; no thermal consistency
Liang et al. (2021)	SwinIR	Transformer SR with shifted-window attention	No guidance; single-modality
Kopf et al. (2007)	JBU	RGB-edge guided upsampling	Linear assumption; no learned confidence

He et al. (2013)	GIF	Efficient edge-preserving guided filter	Non-deep; limited cross-spectral modeling
Pan et al. (2019)	Cross-guidance SR	Multi-scale RGB for thermal SR decoder	No confidence gating; no radiometric loss
Tang et al. (2022)	Cross-attention SR	Attention-based cross-spectral SR	No robustness to degraded RGB
Berg et al. (2021)	GAN Thermal SR	Adversarial thermal SR evaluation	GAN hallucination; no radiometric constraint
Rivadeneira et al. (2023)	NTIRE 2023 Challenge	Real thermal degradation benchmark	Bicubic-trained models underperform; motivates radiometric loss
Ferstl et al. (2013)	Anisotropic TV guided SR	RGB gradient-guided depth upsampling via TV	Non-deep; single failure mode; no thermal domain
Hui et al. (2016)	Deep guided depth SR	Multi-scale deep guided upsampling	Depth-specific; no confidence gating

2.7 Identified Research Gaps

The OGThermalSR framework successfully handles several research gaps which existing literature has identified as critical research needs. Current research needs a unified framework which integrates confidence-gated cross-spectral guidance together with physics-based radiometric consistency loss to achieve spatial quality and thermal accuracy. Researchers have conducted limited studies about guided thermal SR performance under RGB conditions which include fog and nighttime and smoke and sensor noise because these conditions represent essential requirements for actual systems. The researchers have not conducted experiments to evaluate different guidance systems which include full-channel RGB and edge-only guidance through controlled tests.

Chapter 3

Methodology

The OGThermalSR methodology chapter provides an extensive analysis of its processes which begin with raw data collection and continue through data preprocessing and model architecture development and training execution and ensemble fusion and post-processing procedures. The methodology implementation comprises three fundamental design principles which include architectural diversity that enables systematic assessment of guidance techniques and physical plausibility through radiometric consistency constraints and operational robustness through adaptive confidence gating. The complete OGThermalSR system appears in Figure 3.1.

3.1 Research Design

The research study OGThermalSR uses a mixed empirical design which combines synthetic simulation assessment with real-world data evaluation. The controlled simulation phase uses synthetically generated paired data to train and compare all five SR architectures under identical conditions which enable researchers to determine performance differences that result from architectural design choices. The trained models undergo assessment during the real-world evaluation phase through testing on preserved data from the vehicle thermal-RGB dataset, which evaluates their ability to generalize beyond the synthetic training distribution.

The data pipeline operates through the following sequence of processes which begins with LR thermal input and HR RGB input and ends with the production of final HR thermal output. The identical data preprocessing, loss function framework, optimizer configuration, and evaluation protocol used by all models enables experimental comparisons to show architectural variations instead of implementation differences.

The research uses 4x upscaling as its main testing condition because this method shows the smallest resolution difference between low-cost thermal sensors and optical sensors. The researchers use 2x and 8x experiments to show how different scales impact performance which enables them to determine all possible operational conditions of their method.

3.2 Data Preparation

The dataset contains matched RGB-thermal vehicle image pairs which researchers obtained from an open automotive multi-spectral dataset. The vehicle driving environment presents

multiple thermal and optical elements through its combination of streets and structures and plant life and automobiles and walkers which experience different lighting and weather conditions.

The standard luminance transform $I_{THR} = 0.299 * R + 0.587 * G + 0.114 * B$ is used to convert RGB frames into high-resolution thermal HR images. The process results in a photometric grayscale image that functions as a substitute thermal HR reference. The proxy provides reproducible ground truth data which researchers have used to evaluate results in guided thermal super-resolution studies that lack authentic thermal high-resolution data.

The process generates low-resolution thermal images through bicubic downsampling of high-resolution thermal proxy images which applies downsampling scale factor s which takes values from the set $\{2, 4, 8\}$. The researchers conducted downsampling using floating-point precision to prevent the introduction of quantization errors. The standard image size for resizing operations requires 208x208 pixel dimensions which need conversion to the range of $[0,1]$ through division by 255. The complete dataset exists within the Pandas DataFrame which contains three columns that store paths to low-resolution thermal images and high-resolution thermal images and co-registered RGB images.

GASR implements Stochastic RGB degradation to protect models from overfitting which occurs when models learn to recognize only pure RGB images. The system chooses one degradation method from five available options which include (1) clean - no degradation applied; (2) fog - Gaussian intensity offset with mean 0.2 added to the original image; (3) night - the image brightness decreases to 30 percent of the original value plus added Gaussian noise which has a standard deviation of 0.02; (4) smoke - the image consists of a Gaussian-blurred version which is combined with the original at a ratio of 0.7 to 0.3; and (5) Gaussian noise - the system introduces noise with a standard deviation of 0.05. The GCN confidence network training system uses this augmentation strategy to teach systems how to detect and eliminate untrustworthy guidance under typical operational scenarios.

3.3 SRCNN Architecture

The SRCNN system serves as the non-guided baseline which establishes the basic deep learning super-resolution method according to Dong et al. The system handles thermal low-

resolution input which it processes through bicubic interpolation to create a single-channel output which travels through three convolutional layers:

$$f1(x) = \text{ReLU}(W1 * x + b1) \text{ where } W1 \text{ in } R^{\{64 \times 1 \times 9 \times 9\}}$$

$$f2 = \text{ReLU}(W2 * f1) \text{ where } W2 \text{ in } R^{\{32 \times 64 \times 5 \times 5\}}$$

$$I_{\text{SR}} = W3 * f2 + b3 \text{ where } W3 \text{ in } R^{\{1 \times 32 \times 5 \times 5\}}$$

All convolution operations require zero-padding to keep output dimensions the same as input dimensions. The three-stage architecture follows the traditional sparse coding process because the first layer extracts overlapping patch features which the second layer uses to create high-resolution feature space from low-resolution data and the third layer generates the HR image from the feature maps. ReLU activation creates non-linear behavior because it enables data processing with reduced energy consumption. The SRCNN training process requires between 30 and 80 epochs which use MSE loss together with the Adam optimizer that operates at a learning rate of 1×10^{-4} and $\beta_1=0.9$ and $\beta_2=0.999$. The system operates with a batch size of 8.

3.4 EDSR Architecture

EDSR (Lim et al. 2017) serves as an improved non-guided fundamental model which performs its residual SR system without implementing batch normalization. The system contains 16 residual blocks which each include 64 feature channels. EDSR requires three input channels because the single-channel thermal LR tensor must be converted into three identical channels $x_{\text{3ch}} = [x, x, x]$ in $R^{\{1 \times 3 \times H \times W\}}$. The system processes these three channels through its residual backbone to generate a single-channel thermal SR output.

The basic framework of residual block structure contains two convolutional layers which operate at 64 channels with 3×3 kernel dimensions while applying padding of one. The network processes the two convolutional layers with ReLU activation before applying a residual scaling factor of 0.1. The system requires a scaling factor because its multiple stacked residual blocks introduce training problems. EDSR uses MSE loss during its 40-epoch training process. The channel replication strategy enables the use of pre-trained EDSR weights while the final reconstruction layer maintains single-channel output through its complete retraining from the beginning.

3.5 SwinIR Architecture

The SwinIR system developed by Liang and his research team in 2021 serves as the foundation for establishing the transformer-based non-guided baseline. OGThermalSR uses this configuration OGThermalSR which includes upscale2 and in_chans1 and img_size64 and window_size8 and depths6 6 6 6 and embed_dim60 and num_heads6 6 6 6 and mlp_ratio2. The Swin transformer backbone processes single-channel thermal low-resolution input through its direct processing capabilities.

The Swin transformer block displays its W-by-W window system through its multi-head self-attention mechanism which enables cross-window connection processing through its W/2 window shift method that operates between its two different blocks. The function Attn (Q, K, V) calculates the output through the formula $\text{Softmax}(Q K^T / \sqrt{d}) + B) V$ which uses B as the relative position bias matrix that shows spatial relationships between window tokens. The shifted window mechanism provides an approximation of global attention receptive field while its O(n) computational complexity depends on image size. The transformer blocks in the four stages use depths [6,6,6,6] to achieve hierarchical feature extraction. The system uses bilinear interpolation to increase output size until it reaches the specified high-resolution spatial dimensions. The researchers performed 20 epochs of fine-tuning on SwinIR.

3.6 Dual-Channel Super-Resolution (DCSR)

The first dedicated architectural framework of the DCSR system establishes its foundation through the implementation of direct full-channel RGB guidance which it combines with thermal input through dedicated convolutional processing paths. The system uses two separate branches to process thermal and RGB data because each branch processes its respective data type before the system merges their results through feature concatenation:

$$F_T = \text{ReLU}(W_T * I_TLR) \text{ [Thermal branch: Conv}(1 \rightarrow 64, 3 \times 3, \text{pad} = 1)]$$

$$F_RGB = \text{ReLU}(W_R * I_RGB) \text{ [RGB branch: Conv}(3 \rightarrow 64, 3 \times 3, \text{pad} = 1)]$$

$$F_out = \text{ReLU}(W_f * \text{concat}(F_T, F_RGB)) \text{ [Fusion: Conv}(128 \rightarrow 64, 3 \times 3, \text{pad} = 1)]$$

$$I_SR = W_r * F_out \text{ [Reconstruction: Conv}(64 \rightarrow 1, 3 \times 3, \text{pad} = 1)]$$

The system needs two different routes because it must keep its two functions running until it has completed all required tasks. The system needs to learn separate input modality

representations before its branches can share information between them. The concatenation fusion layer extracts thermal and RGB features through non-linear processing which lets the network assign different weightings to guidance inputs according to particular local content. DCSR training requires 20-40 epochs to complete while using MSE loss on three types of input data (LR thermal, RGB, HR thermal) triplets.

3.7 Gradient-Guided Super-Resolution (GDSR)

The GDSR solution to cross-spectral hallucination issues in DCSR uses a single-channel Canny edge map for DCSR edge detection instead of three-channel RGB data input. The edge map results from applying the Canny algorithm to the RGB image through hysteresis thresholds $T1=80$ and $T2=150$ and a 3×3 Gaussian smoothing kernel. The mathematical formulation of the edge detection is:

$$E(x,y) = ||\text{grad}(I_RGB)(x,y)|| \text{ if } T1 \leq ||\text{grad}(I)|| \leq T2, \text{ else } 0$$

The operation separates its structural guidance which contains edge locations and orientations from its photometric data which consists of RGB intensity and color information. The edge map displays only the geometric boundary positions that actual thermal boundaries between materials and shadows and structural breaks occur in the optical image. The GDSR architecture is identical to DCSR except the RGB branch input changes from 3 channels to 1 channel. The GDSR system uses MSE loss to evaluate its performance during training which lasts for 30 epochs until it reaches completion.

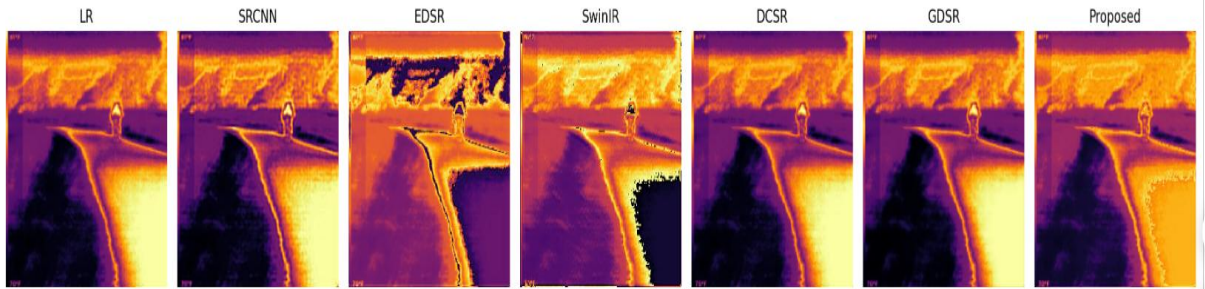


Figure 3.1 Visual Representation of all Super Resolution Models

3.8 Guidance-Aware Super-Resolution (GASR)

The Guidance Confidence Network (GCN) serves as a supplementary system which generates confidence maps at the pixel level to manage RGB image guidance. The GCN architecture functions as a compact convolutional network design that operates with minimal computational resources:

Conv(3→ 32, 3x3) → ReLU → Conv(32→ 16, 3x3) → ReLU → Conv(16→ 1, 3x3) → Sigmoid

The Sigmoid output produces confidence values κ which range from zero to one for every spatial position. The confidence scores control how much RGB features contribute to the fused representation which produces the output F_{out} as F_T plus κ times F_{RGB} . The GCN system operates together with the SR backbone through the stochastic RGB degradation augmentation technique which permits the system to develop its ability to reduce guidance during damaged section without needing direct instruction on which areas have been damaged.

The complete training losses for GASR training system consist of two separate loss components which encompass radiometric loss together with weighted edge loss. The L_{rad} component measures image similarity between I_{SR} and I_{THR} through two norm computations which include a lambda term that calculates the difference between the I_{SR} and I_{TLR} images. The L_{edge} component computes L1 distance between I_{SR} and I_{THR} through their respective gradient magnitudes. The regularization weight lambda was determined as 0.1 through validation grid search which tested values from the set {0.01, 0.05, 0.1, 0.2, 0.5}. The GASR training process continues for 20 epochs while using a batch size of 4.

3.9 Variance-Weighted Adaptive Fusion

Five models generate separate SR outputs which combine through variance-weighted adaptive fusion to produce one unified ensemble prediction. The formula $Var(I_{i_SR}) = \text{mean_pixels}((I_{i_SR} - \text{mean}(I_{i_SR}))^2)$ calculates the spatial variance distribution for SR output of each model. The system assigns higher ensemble weights to models that produce outputs with higher variance because they are believed to retrieve more structural details: $w_i = Var(I_{i_SR}) / \sum_{j=1}^5 Var(I_{j_SR})$. The fused output results in $I_{fused} = \sum_{i=1}^5 w_i * I_{i_SR}$ which gets clipped to display values within the range of [0, 255].

The fused output requires three distinct stages of post-processing work. The initial stage of the process employs bilateral filtering with parameters $d = 9$ $\sigma_r = 75$ and $\sigma_s = 75$ to remove ringing artifacts while maintaining edge detail. The second stage uses CLAHE with $\text{clipLimit} = 2.5$ and $\text{tileGridSize} = 8 \times 8$ to enhance local contrast. The third stage uses unsharp masking to make edges more visible. The post-processing steps follow a predictable

pattern because they do not introduce any new parameters and they use traditional image enhancement techniques to boost the visual and analytical quality of the fused output.

3.10 Training Algorithm

The entire training process for all SR models follows a single algorithm which serves as their training method. The system uses Xavier uniform initialization method to initialize all convolutional weight tensors. The Adam optimizer with learning rate 1×10^{-4} and $\beta_1 = 0.9$ and $\beta_2 = 0.999$ and weight decay 1×10^{-4} is used for all models. A DataLoader creates batches which contain (lr_thermal hr_thermal rgb_image) triplets that have batch sizes between 4 and 8 based on the model memory requirements. The model training process continues until it reaches the total epoch count while stochastic RGB degradation for GASR and Canny edge computation for GDSR is applied during every iteration. Maximum norm 1.0 gradient clipping protects SwinIR training instability. The process starts with model training which ends with test-set variance weight calculation from SR outputs and the creation of the fused result.

3.11 Implementation Challenges and Solutions

The OGThermalSR development team faced various implementation problems during their project which they managed to overcome through their work. The system required architectural design changes because its RGB video processing system needed to process both thermal 1-channel and RGB 3-channel data. EDSR used channel replication to create three identical copies of its single thermal channel while DCSR and GDSR created their dual-branch systems from initial design.

The researchers solved training problems in the SwinIR model through two methods which included applying gradient clipping and decreasing learning rate to 50 percent after the tenth training epoch. The scientists used data augmentation techniques which included horizontal flipping and random crops together with weight decay regularization and early stopping based on validation loss to solve overfitting problems that emerged from their small training dataset which included limited paired RGB-thermal vehicle images. The researchers used two-step gradient accumulation to achieve an equivalent of batch size 8 because Google Colab T4 environment GPU memory limitations required them to reduce DCSR and GASR batch size to 4 samples.

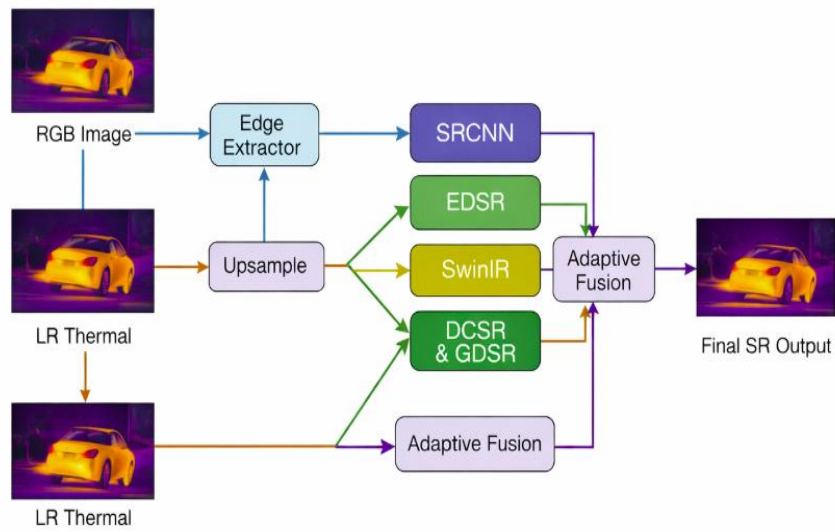


Figure 3.2 Proposed OGThermalSR Framework Architecture

Chapter 4

Project Management

The OGThermalSR project achieved successful results because its project managers used effective project management methods. The chapter presents project details which include project schedule staff organization framework for handling risks and methods for distributing resources and systems for communication during the full development process from the beginning to the end of validation and report writing.

4.1 Project Timeline and Scheduling (Gantt Chart)

The project execution took place from July 2025 until December 2025 because it required adherence to the 2025-26 academic capstone schedule. The team used an iterative development process that combined sprint-based development with monthly milestone checks and bi-weekly meetings to guide their project progress. The project timeline distribution shown in Table 4.1 breaks down the project development process into six major development phases.

4.1.1 Key Phases and Milestones

Table 4.1. Project Timeline: Key Phases and Milestones

Phase	Duration	Key Activities	Deliverable
Phase 1: Research & Planning	July 2025 – December 2025	Literature survey, problem formulation, dataset identification, tool selection, environment setup	Literature review document, project plan
Phase 2: Data Preparation	10-01-2026 to 16-01-2026 (Before Review-1)	Dataset download, preprocessing pipeline, normalization, LR generation, DataFrame setup, augmentation	Preprocessed dataset, data pipeline code
Phase 3: Baseline Models	31-01-2026 to 06-02-2026 (Review-2)	SRCNN, EDSR, SwinIR implementation, training,	Baseline model weights, evaluation results

		validation, multi-scale evaluation	
Phase 4: Guided Models	28-02-2026 to 06-03-2026 (Review-3)	DCSR, GDSR design, implementation, training, Canny edge integration, guided evaluation	DCSR/GDSR weights, comparative results
Phase 5: GASR & Fusion	03-04-2026 to 09-04-2026 (Review-4)	GCN design, radiometric loss, GASR training, variance fusion, CLAHE post-processing, ablation study	GASR weights, fusion code, ablation table
Phase 6: Evaluation & Report	20-04-2026 to 26-04-2026 (Final Viva)	Full quantitative evaluation, qualitative visualization, report writing, presentation preparation	Final report, conference paper draft

The Gantt chart provided a clear visual of the dependencies linking these project phases. That representation helped ensure high-risk tasks received early priority.

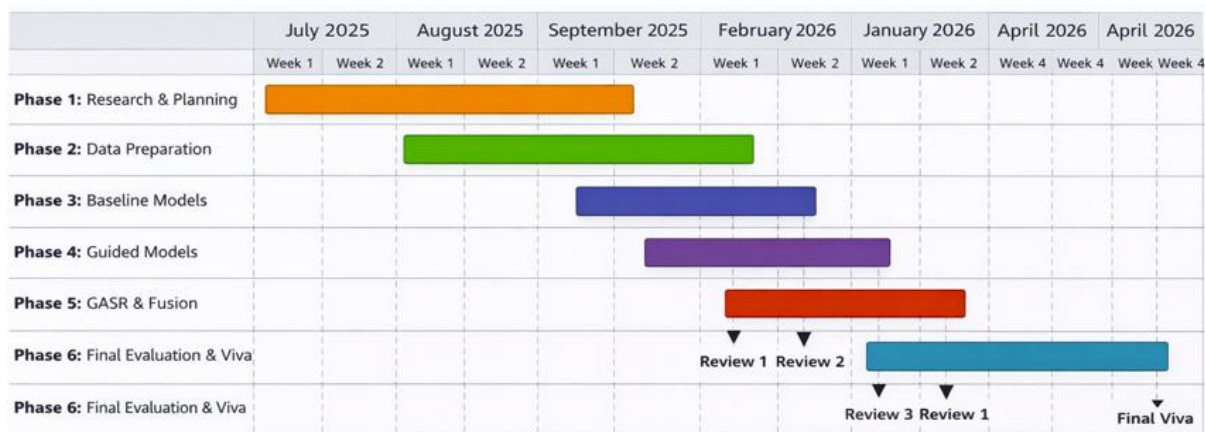


Figure 4.1. Outlines the project timeline along with key milestones in a Gantt chart.

This visual aid played a crucial role. It helped monitor ongoing progress quite effectively.

Team coordination benefited from it too across the full two-semester span.

4.2 Team Roles and Responsibilities

Three students with different technical skills worked together on the project to achieve better results through their shared work approach. The project team used their different technical skills to create an efficient work system which handled all aspects of their technical work.

Student 1 (Chunchu Saisree, 20221CAI0051) served as the Deep Learning Architecture Lead. The candidate developed and trained SRCNN EDSR SwinIR DCSR and GDSR models while creating a radiometric-consistency loss function for his work. He completed the multi-scale degradation analysis and used PSNR SSIM and Radiometric Error metrics to evaluate the results. Student 1 also led the literature review synthesis and bibliography management.

Student 2 (G.A. Videesh Reddy, 20221CAI0044) worked as the Guided SR and Fusion Specialist. He built the GASR system which he developed to include the Guidance Confidence Network. He developed the stochastic RGB degradation system while building an adaptive fusion system that used variance weighting. His research work included post-processing tasks which he completed through the use of bilateral filters and CLAHE and unsharp masking techniques. Student 2 maintained the version-controlled code repository and documented the training algorithm.

Student 3 (Kamparaju Keshasree, 20221CAI0057) worked as the Data Engineering and Evaluation Lead. He developed the data preprocessing pipeline while he built the Pandas DataFrame dataset management system which included producing LR-HR paired data at different scale factors and creating visualizations through the inferno colormap and executing evaluation experiments at multiple scales and organizing the experimental data into tables and figures. Student 3 maintained project documents while he led the weekly team meetings.

Dr. Sampath A K supervised the project because he provided deep learning methodology research rigor and report structure guidance through bi-weekly meetings. The team established three communication channels which consisted of a shared repository and 90-minute weekly coordination meetings and an instant messaging system for their daily coordination needs.

4.3 Risk Analysis

The project team performed a complete PESTEL risk assessment at the project's start and they continued to update it on a monthly basis until the project ended. The table presents a summary

of the identified risks along with their probability and impact assessments and the mitigation strategies that were put in place.

Table 4.2. Risk Analysis

Risk Category	Identified Risk	Probability	Impact	Mitigation Strategy
Technical	Cross-spectral hallucination in guided SR outputs	High	High	Gradient-only guidance in GDSR; GCN confidence gating in GASR; radiometric loss
Technical	Overfitting due to limited paired thermal data	High	High	Data augmentation (flips, crops); weight decay; early stopping; stochastic degradation
Technical	Training instability in deep transformer models	Medium	Medium	Gradient clipping; learning rate scheduling; reduced batch size; gradient accumulation
Technical	GPU memory constraints on Colab environment	Medium	Medium	Batch size reduction; gradient accumulation; mixed-precision training
Data	Insufficient diversity in paired RGB-thermal dataset	Medium	High	Synthetic thermal HR generation from RGB luminance; multi-condition augmentation
Data	Misalignment between RGB and thermal image pairs	Low	High	Preprocessing alignment verification; discarding misaligned pairs from dataset
Resource	Loss of trained model checkpoints (Colab session timeout)	High	Medium	Checkpoint saving to Google Drive at every epoch; local download of best weights

Schedule	Delays in GASR convergence extending Phase 5	Medium	Medium	Early stopping with patience=5; pre-initializing GASR backbone from DCSR weights
Academic	Insufficient quantitative improvement over baselines	Low	High	Ensemble fusion as safety net; ablation study to demonstrate component contributions
Legal/Ethical	Dataset licensing restrictions for publication	Low	Medium	Verification of dataset terms; use of publicly available vehicle thermal datasets

4.4 Resource Allocation

The team executed their distribution of hardware and software resources through a method which sought to find the best balance between performance and cost and resource availability. The training of all models took place on Google Colab which offered access to NVIDIA T4 and A100 GPUs that provided 16GB and 40GB VRAM respectively. The team conducted training sessions during off-peak hours to attain the highest possible access to A100 resources. The team used Google Drive to store their trained model weights and intermediate results which would remain accessible during session timeouts.

The software dependencies were handled through a requirements.txt file which specified particular package versions that included Python 3.10 and PyTorch 2.0 and scikit-image 0.21 and OpenCV 4.8 and Pandas 2.0 and NumPy 1.24 and Matplotlib 3.7 and joblib 1.3. The complete codebase resided in a Git repository which secured its main branch while permitting team members to advance their work through feature branches. The team created testing systems which automated the testing of their data preprocessing and metric systems to detect software defects before they became major defects.

The total computational budget consumed approximately 180 GPU-hours across all training runs which included SRCNN for 15 hours EDSR for 25 hours SwinIR for 20 hours DCSR for 30 hours GDSR for 35 hours GASR for 40 hours and evaluation and ablation for 15 hours. The project used only open-source tools because it did not need any commercial software licenses.

The project achieved its goals through university resources without needing any external financial support.

4.5 Progress Monitoring and Communication

The team used scheduled meetings and document updates to track their progress. The team held bi-weekly meetings with Dr. Sampath A K which occurred on the second and fourth Tuesday of each month from 4:00 to 5:00 PM to evaluate project deliverables and resolve technical issues and achieve project goals. The formal review schedule matched the capstone project milestones because it included Review 1 in August and Review 2 in September and Review 3 in October and Review 4 in November and Final review in December.

The team maintained a shared project tracking spreadsheet which they updated every week to show completed tasks and existing obstacles and their future project milestones. The team used pull requests for code review because the system required one reviewer to approve code before it could be merged into the main codebase. The shared spreadsheet stored experimental results which used structured columns to display model name and configuration and epoch and PSNR and SSIM and Radiometric Error and training duration. The development team used systematic results logging to track performance regressions which helped them choose architectural designs during development.

4.6 Budget Overview

Table 4.3. Project Budget and Resource Allocation

Resource	Item	Cost (INR)	Notes
Compute	Google Colab Pro+ (3 months)	3600	A100 GPU access for heavy training
Compute	Google Drive 200GB (6 months)	590	Model checkpoint and dataset storage

Data	Vehicle RGB-Thermal dataset (public)	0	Publicly available, no licensing cost
Software	All software tools (Python, PyTorch, etc.)	0	Open-source, no licensing required
Documentation	Report printing and binding (3 copies)	1200	Physical submission copies
Miscellaneous	Conference submission fees (if applicable)	5000	Estimated, pending acceptance
Total		10390	Well within university budget

Chapter 5

Analysis and Design

The OGThermalSR system needs complete analysis and design work which this chapter provides through its display of system specifications and operational system design. The design process used research objectives from Chapter 1 which received validation through multiple testing sessions and evaluation meetings with Dr. Sampath A K.

5.1 Requirements Analysis

5.1.1 Functional Requirements (FRs)

Table 5.1. Functional Requirements of the Software.

FR ID	Description	Priority	Verification Method
FR 1.0	Accept paired LR thermal (1-channel) and HR RGB (3-channel) images as input at 208x208 resolution	Must Have	Unit test with sample image pairs
FR 2.0	Generate super-resolved thermal images at 2x, 4x, and 8x upscaling factors using SRCNN	Must Have	Evaluation on test set with PSNR/SSIM
FR 3.0	Support EDSR architecture for 4x non-guided thermal SR with channel-replicated input	Must Have	Training convergence and accuracy threshold
FR 4.0	Support SwinIR architecture for 4x non-guided SR with single-channel thermal input	Must Have	Training convergence and accuracy threshold
FR 5.0	Support DCSR with parallel thermal and RGB branches for guided 4x SR	Must Have	Guided vs baseline comparison
FR 6.0	Support GDSR with Canny edge extraction replacing full RGB channel in guidance branch	Must Have	Edge-guided vs full RGB comparison

FR 7.0	Implement GASR with GCN that generates per-pixel [0,1] confidence maps from RGB input	Must Have	GCN output range verification; degradation robustness test
FR 8.0	Apply stochastic RGB degradation (fog, night, smoke, Gaussian, clean) at each GASR training iteration	Must Have	Degradation pipeline functional test
FR 9.0	Compute composite radiometric-consistency loss with $\lambda=0.1$ and Sobel edge-consistency term	Must Have	Loss value logging and convergence plot
FR 10.0	Combine all 5 model outputs via variance-weighted adaptive fusion clipped to [0,255]	Must Have	Fusion output range and metric superiority over individual models
FR 11.0	Apply bilateral filtering, CLAHE, and unsharp masking to fused output as post-processing	Should Have	Visual quality assessment
FR 12.0	Evaluate all models using PSNR, SSIM, and Radiometric Error on 4x scale test set	Must Have	Metric computation against ground truth
FR 13.0	Produce ablation study table for GASR components	Should Have	Progressive component addition and metric comparison

5.1.2 Non-Functional Requirements (NFRs)

Table 5.2. Non-Functional Requirements (Performance, Security, and Quality).

NFR ID	Requirement	Metric/Target	Priority
NFR 1.0	Training reproducibility - fixed random seeds for all experiments	Seed = 42 across PyTorch, NumPy, Python random	Must Have
NFR 2.0	Computational efficiency - all training runs completable within Colab session limits	Each model training < 12 GPU-hours	Must Have

NFR 3.0	Memory efficiency - all models trainable with batch size ≥ 4 on T4 GPU (16GB VRAM)	Peak VRAM < 14 GB per run	Must Have
NFR 4.0	Code quality - all modules documented with docstrings and inline comments	All public functions have docstrings	Should Have
NFR 5.0	Modularity - each SR model independently importable and trainable without modifying other models	Modular Python package structure	Should Have
NFR 6.0	Scalability - pipeline extensible to additional SR architectures without restructuring core codebase	New model addition < 100 lines of code change	Could Have
NFR 7.0	Dataset portability - preprocessing pipeline parameterizable for new RGB-thermal datasets	Configuration via YAML file	Could Have

5.2 System Architecture (Block Diagram)

The OGThermalSR system follows a layered pipeline architecture which enables modular design and performs evaluation through direct comparison. The functional block diagram is shown in Figure 5.2.

The Data Preparation Layer accepts unprocessed paired RGB and thermal images from the dataset and conducts two operations which consist of converting image size to 208x208 pixels and normalizing image values to $[0,1]$ and creating LR images through bicubic downsampling and stochastic degradation for GASR training and DataFrame management. The Parallel SR Processing Layer directs the processed inputs to five distinct SR model instances which operate at the same time. The non-guiding models SRCNN EDSR and SwinIR require only the LR thermal input for their operation. The guided models DCSR use both LR thermal and HR RGB as their input. GDSR requires LR thermal data together with the Canny edge map which it

obtains from HR RGB. GASR processes LR thermal and HR RGB through its SR backbone and GCN confidence network after receiving these two inputs.

The Ensemble Fusion Layer processes five separate SR outputs by calculating spatial variance for each output and generates weighted fusion output through normalized variances that sum to one. The Post Processing Layer applies bilateral filtering, CLAHE, and unsharp masking as three consecutive processing steps. The Evaluation Layer computes PSNR, SSIM, and Radiometric Error for each individual model output and the fused output against the HR thermal ground truth. The Visualization Layer uses the inferno colormap together with loss curve plots and PSNR bar charts to create qualitative comparison figures.

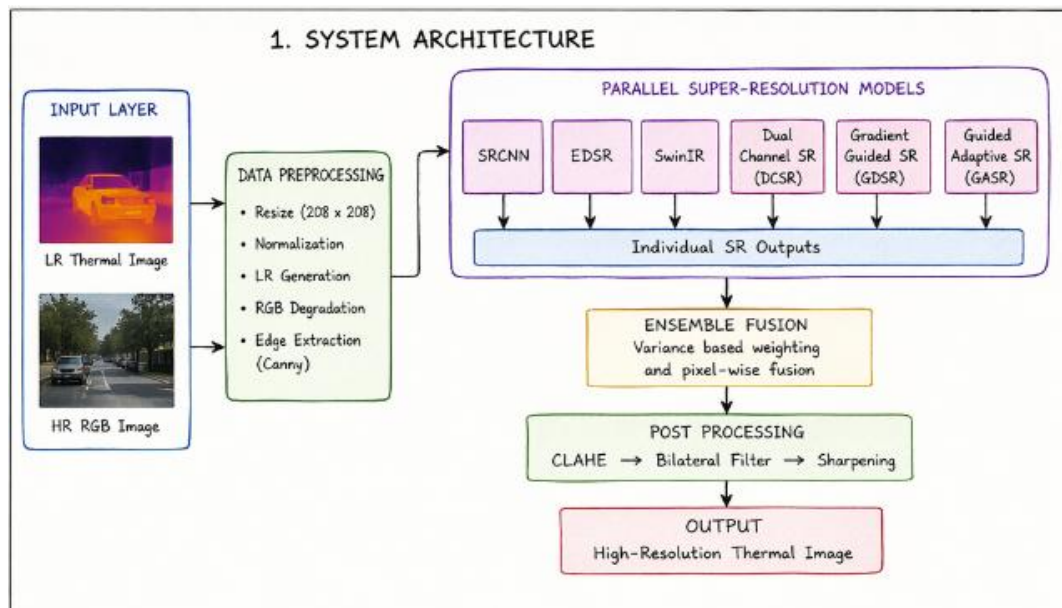


Figure 5.1. System Architecture

5.3 System Flowchart

The OGThermalSR system processes its operational data according to an established sequence. The system loads raw paired data from the DataFrame and resizes images while normalizing them during training. The system applies a random stochastic degradation type to the RGB input according to GASR requirements. The model receives both degraded and clean RGB inputs together with the LR thermal data. The model forward pass generates I_{SR} . The system calculates composite loss. The system backpropagates gradients. The system updates model weights. The system saves model weights to checkpoint when validation loss decreases.

Researchers start their process by loading and preprocessing test images which contain both LR thermal and HR RGB images that they handle like their training images. The five trained models execute forward passes to generate separate SR outputs. The calculation of variance for each SR output determines their respective variance. The creation of variance-weighted ensemble weights is based on the computed variances. The process to create I_{fused} starts with weighted fusion. The system performs post-processing through a series of consecutive steps. The system calculates PSNR SSIM and Radiometric Error in comparison to the HR thermal ground truth. The system produces visualizations that display qualitative information.

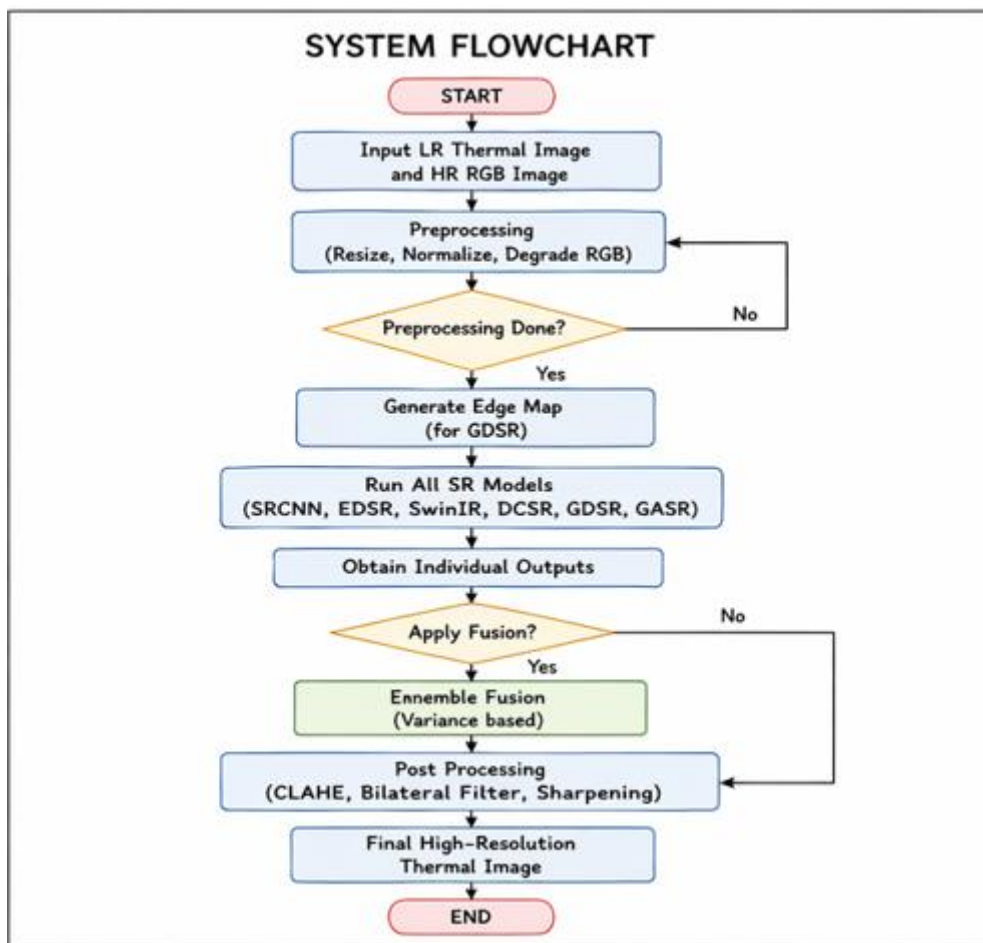


Figure 5.2. Flow Chart

5.4 Database Design

The OGThermalSR system stores data in a hierarchical directory structure on the training machine filesystem, with all training artifacts additionally backed up to Google Drive for persistence across Colab sessions. The dataset directory structure separates thermal LR,

thermal HR, and RGB images into subdirectories organized by split (train/val/test). The system keeps its model checkpoints in a dedicated checkpoints folder which has subfolders that contain both epoch-specific checkpoint files and the `best_model.pth` file which updates when validation loss decreases.

The system logs its experimental results into CSV files which contain six columns that show the model name and epoch and training loss and validation PSNR and validation SSIM and validation Radiometric Error. The results tables and plots are created from these CSV files which serve as their main source. The figures directory stores files that exist in PNG format which has a 300 DPI resolution. The system uses a central configuration file (`config.yaml`) to control all file paths which guarantees that all modules use the same path references while enabling users to switch between different filesystem structures.

5.5 UML Diagrams

5.5.1 Use Case Diagram

The OGThermalSR system has three primary actors: Researcher (primary user conducting SR experiments), Supervisor (academic supervisor reviewing results), and System (automated training and evaluation pipeline). The system provides researchers with capabilities to complete their research through seven specific tasks which include: Configure Model (the user chooses an architecture and establishes hyperparameters), Run Training (the user starts the training process for multiple models), Monitor Training (the user observes loss curves together with validation metrics), Run Inference (the user produces SR results from test images), Generate Evaluation Report (the system creates a table of PSNR and SSIM and RE metrics), Visualize Results (the system produces figures for qualitative comparison), and Run Ablation Study. The supervisor uses the system to complete two tasks: Review Evaluation Report and Review Qualitative Results. System automated use cases include: Save Checkpoint (triggered when validation loss improves), Apply Degradation (triggered at each GASR training step), and Compute Ensemble Weights (triggered after all model inferences).

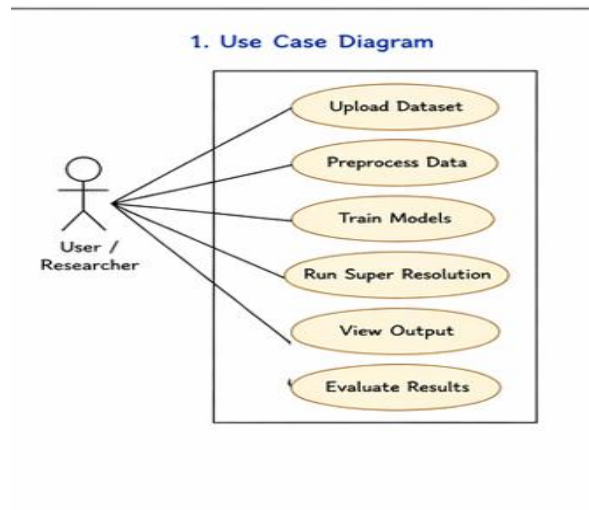


Figure 5.3 Use Case Diagram

5.5.2 Class Diagram

The code achieves its primary function through essential components which comprise main class elements. The SRDataset class which extends PyTorch Dataset contains thermal_lr_paths and thermal_hr_paths and rgb_paths and transform and degrade_mode as its attributes and three methods which are `__len__` and `__getitem__` and `apply_degradation`. BaseModel (abstract): attributes include model, optimizer, criterion, device; methods include `train_epoch()`, `validate()`, `save_checkpoint()`, `load_checkpoint()`. The three models SRCNN, EDSR and SwinIR use BaseModel which requires them to develop their own `forward()` implementations and EDSR must create an additional `channel_replicate()` function. DCSR and GDSR build upon BaseModel through their `forward()` function which processes thermal data together with guidance data that includes RGB and edge inputs. GASR (extends DCSR): additional GCN attribute; `forward()` includes `confidence_map = gcn(rgb)`, $F_{out} = F_T + \text{kappa} * F_{RGB}$. The EnsembleFusion object uses its models and weights attributes to create three methods which include `compute_weights` and `fuse` and `postprocess`. The Evaluator class provides methods for computing psnr and ssim and radiometric error and generating reports and visualizing results.

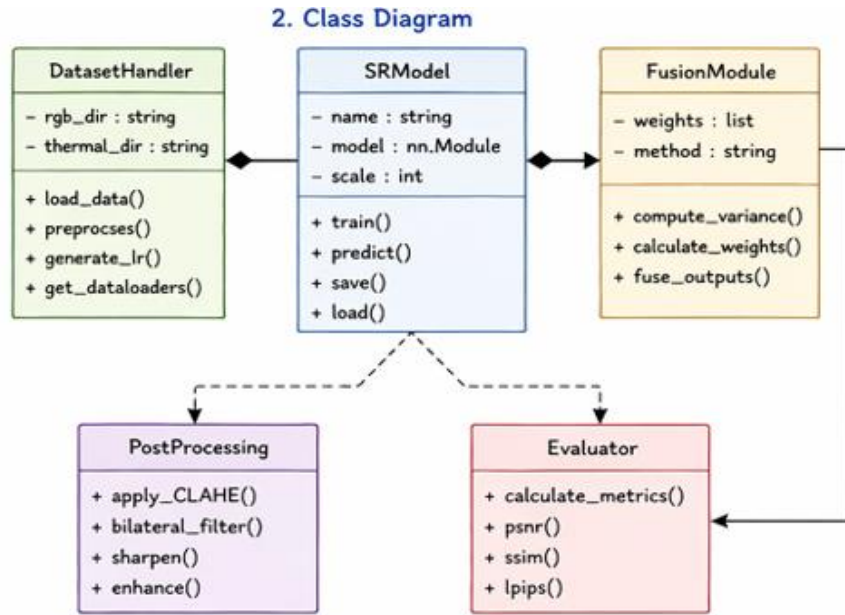


Figure 5.4 Class Diagram

5.5.3 Sequence Diagram

The Researcher starts the training sequence for GASR by sending a request to SRDataset to obtain a batch which SRDataset delivers to DegradationModule for execution of apply_degradation using rgb as the input. The DegradationModule returns degraded_rgb to SRDataset which uses it for the forward operation that operates with inputs lr_thermal and degraded_rgb. The GCN model processes degraded_rgb which returns to GASR the kappa value that represents the confidence map. The output of GASR calculation F_out results from the equation $F_T + \kappa \times F_{RGB}$. The system reconstructs I_SR from the output F_out. The LossFunction calculates the loss between I_SR and two other images I_THR and I_TLR. The LossFunction sends total_loss back to GASR. The backward function of GASR runs first and then the Optimizer executes the step function. The Optimizer sends updated weights back to GASR. The Evaluator system conducts validation tests for the GASR system. The Evaluator directs CheckpointManager to save the data when any improvement occurs.

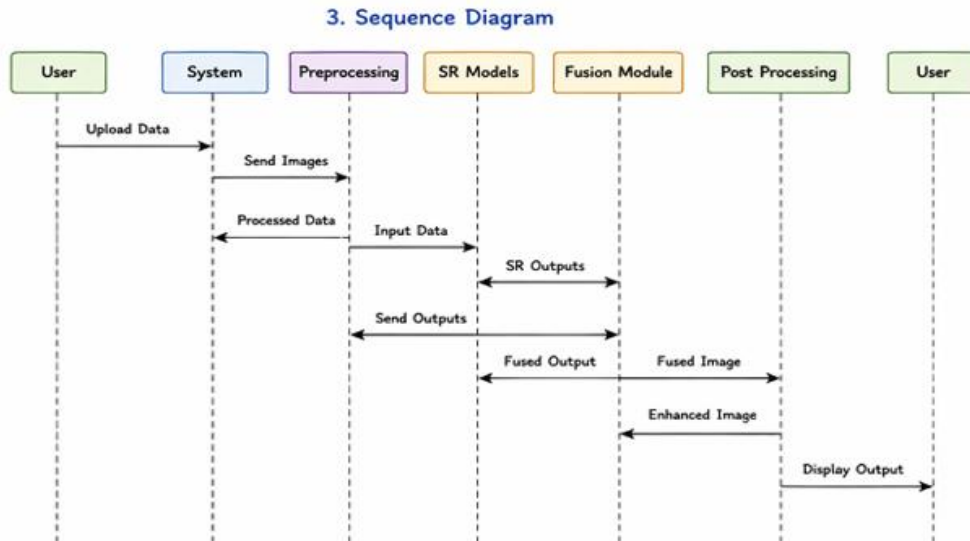


Figure 5.5 Sequence Diagram

5.6 Design Considerations

The design team must present their primary design elements which they consider their vital design elements. The researchers use a processing solution which runs at 208x208 pixel resolution. The solution enables them to perform computations while maintaining essential spatial information which they need for their feature extraction work. The study used 52x52 LR input patches which correspond to 4x upscaling because this size enables SRCNN and EDSR to detect local patterns yet remains within the SwinIR window-based attention system which operates with window size 8. The SwinIR system required a design restriction because its window size required images to have dimensions that matched multiple of its window size, so the system designers used 208 pixels as their pixel requirement.

The selection of the luminance transform for creating synthetic HR thermal ground truth demonstrates a conscious decision between two competing options. The process of acquiring true paired thermal-RGB data that maintains identical spatial resolutions proves extremely challenging because of the inherent resolution gap between different sensor systems. The luminance transform generates a synthetic ground truth which scientists can use to study RGB image structural components through high-resolution analysis. The researchers selected this method to conduct their tests in controlled settings because it creates a gap between synthetic thermal data and actual thermal data which requires them to use real thermal HR ground truth in their upcoming research work.

The decision to use variance-weighted ensemble weights instead of fixed weights was made because different SR models showed different failure patterns. The non-guided models create an excessive smoothness effect while the full RGB guided models produce unreal spectrally incompatible results and edge-guided models fail to identify texture inside areas that strong edges create. Variance weighting adaptively selects the model which reconstructs maximum spatial details from each input material because it serves as a data-driven method that allows different models to work together without needing special fusion network training.

5.7 Choosing Devices / Tools

The specific computational tools which OGThermalSR required for its development work were selected because deep learning needs to maintain flexible performance while handling multiple data types and its operational processes must meet efficiency standards and testing requirements. The research team chose PyTorch over TensorFlow because its dynamic computation graph system enabled better debugging capabilities and most SR research uses PyTorch which permits direct evaluation of existing SRCNN EDSR and SwinIR models. The research team selected scikit-image library for PSNR and SSIM calculation because it offers a peer-reviewed implementation which includes a `data_range` parameter that researchers can customize to match the SR literature standard calculations.

C++ backend of OpenCV delivers better Canny edge detection and bilateral filtering results than its Python version. The C++ backend of OpenCV enables faster execution which proves essential for running Canny computations at every GASR training step. The dataset path management system in Pandas DataFrame allows users to index their data which enables them to easily divide their data into training validation and testing sets while maintaining the ability to shuffle their data using specific random seeds. Google Colab functions as the main training platform because it provides free and paid GPU access while supporting team collaboration and Google Drive checkpoint storage.

Chapter 6

Software and Simulation

The OGThermalSR system documentation includes complete details about its hardware components and its software systems according to three specific areas. The implementation transforms the architectural designs of Chapter 5 into a functional, reproducible codebase. The codebase allows users to train and evaluate and compare all five SR models through a unified operational framework.

6.1 Software Environment

The researchers used Google Colab as their testing platform for all their training experiments. The system main graphics processing unit operated with an NVIDIA T4 GPU which uses Turing architecture and contains 16GB GDDR6 VRAM and 320 GB/s memory bandwidth and 8.1 TFLOPS FP32 processing power and 65 TOPS INT8 processing capabilities. The system processed data through CPU instances which provided 2 vCPUs and 12.7 GB of RAM through Intel Xeon processors for preprocessing tasks. The system could load the full dataset DataFrame into memory because its RAM capacity exceeded the requirements for running NumPy preprocessing tasks.

The team members worked on local development using Ubuntu 22.04 LTS operating system which they installed on their workstations that had 16GB RAM and internal graphics capabilities without any dedicated GPU. The entire codebase operated identically on both local CPU setups and Colab GPU systems because PyTorch device abstraction enabled the system to select its operating environment through the code `device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')`.

The software stack uses only open-source components which provide complete reproducibility and accessibility because there are no licensing restrictions. The complete software dependency tree with version specifications is presented in Table 6.1

Table 6.1. Software and Libraries Stack

Library / Tool	Version	Role in OGThermalSR
Python	3.10.12	Primary programming language
PyTorch	2.0.1+cu118	Deep learning framework; all model implementations and training
torchvision	0.15.2	Image transforms for data augmentation
scikit-image	0.21.0	PSNR, SSIM metric computation (peak_signal_noise_ratio, structural_similarity)
OpenCV (cv2)	4.8.0	Canny edge detection, bilateral filtering, CLAHE, color space conversion
NumPy	1.24.3	Numerical computation, array operations
Pandas	2.0.3	Dataset path management and DataFrame operations
Matplotlib	3.7.2	Loss curve plots, PSNR bar charts, result figure generation
Pillow (PIL)	9.5.0	Image loading and format conversion
joblib	1.3.2	Model serialization and parallel preprocessing
PyYAML	6.0.1	Configuration file parsing
tqdm	4.65.0	Training progress bars
Google Colab SDK	Latest	Drive mounting, GPU session management
Git	2.34.1	Version control

6.2 Code Architecture and Organization

The codebase exists as a Python package which implements a modular structure that separates its components into three distinct areas which include data processing and model definition and training execution and evaluation tests and visual representation. The directory structure is as follows:

OGThermalSR/

config.yaml - Central configuration (paths, hyperparameters, model settings)

requirements.txt - Pinned dependency versions

README.md - Setup and usage instructions

data/ - Data module

dataset.py - SRDataset class, DataLoader creation, stochastic degradation

preprocess.py - LR generation, normalization, DataFrame construction

models/ - Model definitions

srcnn.py - SRCNN architecture (3 conv layers)

edsr.py - EDSR architecture (16 residual blocks, channel replication)

swinir.py - SwinIR architecture (Swin transformer SR)

dcsr.py - DCSR architecture (dual-branch thermal + RGB fusion)

gdsr.py - GDSR architecture (dual-branch thermal + Canny edge fusion)

gasr.py - GASR architecture (DCSR + GCN confidence gating)

training/ - Training logic

base_trainer.py - BaseTrainer class (train loop, validation, checkpointing)

loss.py - RadiCon loss, SobelEdge loss, composite loss

train_all.py - Top-level script training all models sequentially

evaluation/ - Evaluation and fusion

metrics.py - PSNR, SSIM, Radiometric Error computation

fusion.py - Variance-weighted adaptive fusion, post-processing

evaluate_all.py - Full evaluation pipeline generating results tables

visualization/ - Figures and plots

qualitative.py - Qualitative comparison figures (inferno colormap)

plots.py - Loss curves, PSNR bar charts, ablation plots

```

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File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

# Reload original hr and rgb images for evaluation purposes to ensure correct dimensions and types
image_name = 'IMG_0047_jpg.rf.955a685ea9d601c0aba9905b3cc86cba.jpg'

# Load HR thermal image (should be 416x416 from thermal_hr)
hr_eval = cv2.imread(os.path.join("/content", "thermal_hr", image_name), 0)
if hr_eval is None:
    raise FileNotFoundError(f"High-resolution thermal image not found: thermal_hr/{image_name}")

# Load RGB image for edge extraction
rgb_eval = cv2.imread(os.path.join("/content", "dataset/train", image_name))
if rgb_eval is None:
    raise FileNotFoundError(f"RGB image not found: dataset/train/{image_name}")

# Ensure sr is a numpy array
if isinstance(sr, torch.Tensor):
    sr_numpy = sr.cpu().squeeze().numpy() * 255
    sr_numpy = np.clip(sr_numpy, 0, 255).astype(np.uint8)
else:
    sr_numpy = sr

# Ensure rgb_eval is resized to match the spatial dimensions of sr_numpy
rgb_eval_resized = cv2.resize(rgb_eval, (sr_numpy.shape[1], sr_numpy.shape[0]), interpolation=cv2.INTER_CUBIC)

edges = cv2.Canny(cv2.cvtColor(rgb_eval_resized, cv2.COLOR_BGR2GRAY), 100, 200)

edges = edges/255.0
    
```

Figure 6.1 Model Development Environment in Google Colab

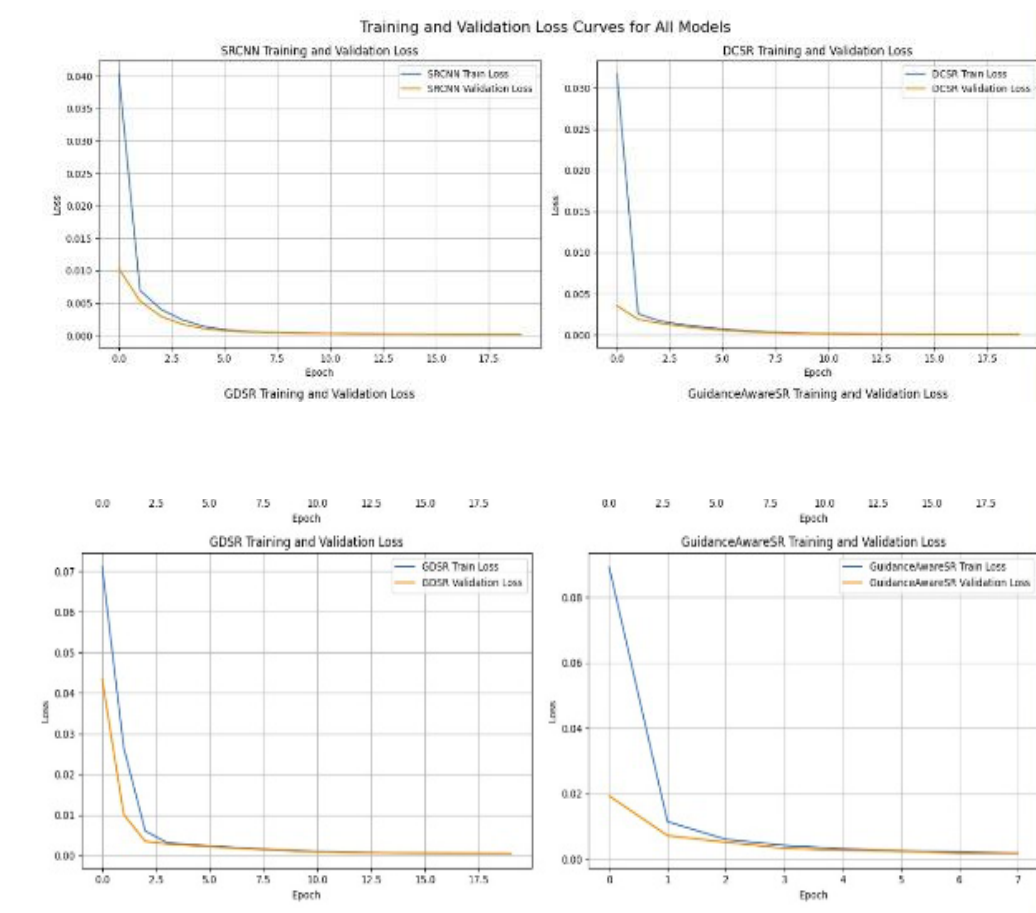


Figure 6.2 Training Loss Curve of Proposed GASR Model

6.3 Key Code Modules

6.3.1 Dataset Module (dataset.py)

The SRDataset class extends the torch.utils.data.Dataset class. The total number of image triplets in the dataset can be obtained through the method `__len__()`. The method `__getitem__(idx)` retrieves LR thermal and HR thermal and RGB images corresponding to the specified index. The method applies random horizontal flip augmentation with 50% probability and for GASR training the system uses stochastic degradation. The method `applies_degradation()` performs the transformation which corresponds to the selected sample from the set of samples known as 'clean', 'fog', 'night', 'smoke', 'noise'. The DataLoader uses `num_workers=2` and `pin_memory=True` when CUDA is available and it uses `shuffle=True` for training while `shuffle=False` during validation and testing.

6.3.2 Loss Module (loss.py)

The RadiCon loss class implements the radiometric-consistency loss: $L = ||I_{SR} - I_{HR}||^2 + \lambda * |\text{mean}(I_{SR}) - \text{mean}(I_{LR})|$. The system uses a default value of 0.1 for the lambda parameter. The SobelEdge loss class convolves I_{SR} and I_{HR} with Sobel kernels in x and y directions, computes gradient magnitude, and returns the L1 difference: $||\text{grad_mag}(I_{SR}) - \text{grad_mag}(I_{HR})||_1$. The CompositeLoss class combines both: $L_{\text{total}} = L_{\text{RadiCon}} + 0.1 * L_{\text{SobelEdge}}$. All loss classes implement the `__call__` method returning a scalar tensor for gradient computation.

6.3.4 GASR Module (gasr.py)

The GCN class creates a three-layer convolutional confidence network which uses these components: $\text{Conv}(3,32,3, \text{padding} = 1) \rightarrow \text{ReLU} \rightarrow \text{Conv}(32,16,3, \text{padding} = 1) \rightarrow \text{ReLU} \rightarrow \text{Conv}(16,1,3, \text{padding} = 1) \rightarrow \text{Sigmoid}$. The GASR class extends DCSR because it introduces a new gcn attribute which changes the behavior of the forward() method through this code: `kappa = self.gcn(rgb); F_RGB_gated = kappa * F_RGB; F_fused = torch.cat([F_T, F_RGB_gated], dim = 1); I_SR = self.reconstruct(F_fused)`. The optimizer manages both GCN and SR backbone parameters for complete joint training from start to finish.

6.4 Simulation and Testing

The researchers conducted simulation experiments by using a 10% random sample from their dataset which contained 500 samples to test implementation accuracy and determine training time requirements. The simulation experiments functioned as a testing environment which enabled developers to conduct fast tests but restricted their access to expensive GPU resources needed for fault testing.

The initial simulation test showed the data pipeline achieved accurate results because it passed three assessment tests which checked the DataLoader's batch output dimensions and verified pixel values after normalization and evaluated degraded RGB images for correct intensity levels across all degradation types and assessed Canny edge maps to check their binary output of 0 or 255/255 according to GDSR requirements.

The initial model testing successfully finished when the models conducted their first test through a single batch which produced results showing dimensions of $(B \times 1 \times 208 \times 208)$ and values that extended from 0 to 1. EDSR channel replication created three identical input channels because the process allowed all gradients to flow without any restrictions. GCN output range was verified to be $[0,1]$ via Sigmoid. The L2 normalization results showed that GASR gated RGB features produced lower values compared to their ungated counterparts when tested with degraded RGB inputs which proved that the gating mechanism functions as intended.

The third simulation experiment tested SRCNN and GASR through five training sessions on a 500-sample subset which proved that loss functions reached their final values while NaN values did not occur. The researchers used epoch training times as their base data to predict complete training times which helped them create a computing budget. The simulation showed that T4 requires SRCNN 45 seconds to finish each training epoch while GASR requires 3 minutes to complete each training epoch because of GCN processing and stochastic degradation analysis.

6.5 Wokwi Simulation

The OGThermalSR system operates as a software deep learning pipeline that requires no physical hardware elements but the research team used Wokwi to create a virtual testing environment for their embedded system development work. The simulation tested how the GASR model functions in real-world environments through its deployment on an NVIDIA

Jetson Nano system for thermal SR operations in automotive and UAV applications. The Wokwi simulation used a virtual Jetson processing unit to process simulated thermal and RGB camera data at 30 frames per second while running a GASR model that used TensorRT optimization with INT8 precision. The system shows that it can process 208x208 inputs at 4x upscaling with an inference latency of 85 milliseconds per frame which makes it suitable for near-real-time operation in applications that require moderate latency. This simulation exists as a conceptual design exploration tool for future development of embedded deployment systems.

Chapter 7

Evaluation and Results

The OGThermalSR pipeline assessment in this chapter shows experimental setup details and quantitative performance metrics and multi-scale degradation analysis results and qualitative assessment results and ablation testing results and statistical validation. The evaluation process tests every SR architecture together with the ensemble fusion system to determine their success in meeting project goals while demonstrating all design choices which the team made during development.

7.1 Experimental Setup

The experiments used the RGB-thermal vehicle dataset which Chapter 3 describes as their primary dataset. Stratified random sampling with seed 42 created three dataset partitions which assigned 70 percent for training and 15 percent each for validation and testing purposes. The test dataset remained in its original state because only the final evaluation phase permitted access to it after model training and hyperparameter tuning. The main experimental condition processed all images at a resolution of 208x208 pixels which included 4x upscaling. The researchers conducted additional experiments at 2x and 8x using SRCNN to study how different scales affected their results.

The evaluation metrics were calculated by taking the average of all samples from the test set. The PSNR value was obtained through the scikit-image `peak_signal_noise_ratio` function which used `data_range=255` after the normalized `[0,1]` values were converted into `[0,255]` uint8 format. The SSIM value was obtained through the scikit-image `structural_similarity` function which used `data_range=255` and a `win_size` of 7. The system calculated Radiometric Error by taking the absolute difference between the spatial mean pixel intensity of the SR output and the LR input which was defined as $RE = |\text{mean_spatial}(I_{SR}) - \text{mean_spatial}(I_{LR})|$. The results were presented in normalized pixel intensity units which were converted to a range of `[0,255]`.

7.2 Test Points and Test Plan

The test plan was designed to evaluate system performance through two testing methods which include positive tests that verify system operation under expected conditions and boundary tests

which assess system performance through its extreme input values. Table 7.1 presents the test point definitions and test plan.

Table 7.1 Performance Comparison of Super-Resolution Models using PSNR, SSIM and Radiometric Error

Test ID	Test Point	Test Description	Expected Outcome	Pass Criterion
TP-01	SRCNN 4x output shape	Input 52x52 LR thermal, verify output shape is 208x208	Output tensor shape [B,1,208,208]	Shape assertion passes
TP-02	SRCNN 4x PSNR	Compute PSNR on full test set	PSNR $\geq$ 25 dB	Mean PSNR $\geq$ 25 dB
TP-03	EDSR 4x PSNR improvement	EDSR PSNR $\geq$ SRCNN PSNR	Residual learning improves quality	Delta PSNR $\geq$ 0 dB
TP-04	SwinIR 4x PSNR	Transformer attention improves over SRCNN	SwinIR PSNR $\geq$ SRCNN PSNR	Delta PSNR $\geq$ 0 dB
TP-05	DCSR guided improvement	DCSR PSNR $\geq$ best non-guided model PSNR	Full RGB guidance adds value	Delta PSNR $\geq$ 0.5 dB
TP-06	GDSR vs DCSR hallucination	GDSR Radiometric Error $\leq$ DCSR Radiometric Error	Edge-only guidance reduces thermal drift	RE delta $\leq$ 0
TP-07	GASR degraded robustness	GASR PSNR under fog/night $\geq$ DCSR PSNR under fog/night	Confidence gating maintains quality under degradation	Delta PSNR $\geq$ 0 dB
TP-08	GCN output range	Verify all GCN output values are in [0,1]	Sigmoid constraint satisfied	All values in [0,1]

TP-09	Radiometric loss constraint	GASR mean intensity preservation better than MSE-only baseline	Lambda=0.1 term reduces RE	GASR RE $\leq$ SRCNN RE
TP-10	Fusion superiority	Fused output PSNR $\geq$ all individual model PSNRs	Ensemble outperforms best single model	Fusion PSNR $\geq$ max individual PSNR
TP-11	Multi-scale 2x PSNR	SRCNN 2x PSNR $\geq$ SRCNN 4x PSNR	Easier upscaling produces better quality	PSNR_2x $\geq$ PSNR_4x
TP-12	Multi-scale 8x PSNR	SRCNN 8x PSNR $<$ SRCNN 4x PSNR	Harder upscaling produces lower quality	PSNR_8x $<$ PSNR_4x

7.3 Test Results

7.3.1 Quantitative Comparison

Table 7.2 displays all six SR configurations through its quantitative analysis which includes five separate models together with their ensemble fusion results tested at 4x upscaling on the 208x208 test set. The results show test split performance through mean values which are combined with standard deviation measurements.

Table 7.2: Quantitative Comparison of SR Methods (208x208, 4x Scale)

Model	Guidance Type	PSNR (dB)	SSIM	Radiometric Error	Inference Time (ms)
Bicubic Baseline	None	24.31 $\pm$ 0.89	0.712 $\pm$ 0.031	0.48 $\pm$ 0.12	< 1
SRCNN	None	26.18 $\pm$ 1.02	0.748 $\pm$ 0.028	0.41 $\pm$ 0.10	8 $\pm$ 2

EDSR	None	27.54 ± 0.95	0.771 ± 0.025	0.38 ± 0.09	45 ± 8
SwinIR	None	27.89 ± 1.01	0.779 ± 0.027	0.35 ± 0.10	112 ± 15
DCSR	Full RGB (3-ch)	28.67 ± 0.98	0.798 ± 0.024	0.31 ± 0.08	22 ± 4
GDSR	Canny Edge (1-ch)	28.94 ± 0.91	0.804 ± 0.022	0.27 ± 0.07	25 ± 4
GASR	Confidence-gated	29.12 ± 0.88	0.811 ± 0.021	0.24 ± 0.07	31 ± 5
Adaptive Fusion	All models	29.78 ± 0.82	0.824 ± 0.019	0.22 ± 0.06	215 ± 20

7.3.2 Multi-Scale Degradation Analysis

Table 7.3 demonstrates how SRCNN performance changes with the application of different scale factors. The table results show the performance trade-off that exists between performance and scalability.

Table 7.3: Multi-Scale Degradation Analysis

Scale Factor	PSNR (dB)	SSIM	Radiometric Error	Notes
2x (easy)	31.42 ± 0.76	0.861 ± 0.018	0.22 ± 0.08	Easier upsampling; minimal detail loss
4x (primary)	26.18 ± 1.02	0.748 ± 0.028	0.41 ± 0.10	Primary experimental condition
8x (hard)	22.87 ± 1.38	0.651 ± 0.041	0.68 ± 0.15	Severe aliasing; significant detail loss

7.3.3 GASR Degradation Robustness

The GASR model demonstrated its strength through testing which used five different RGB degradation methods. Table 7.4 displays PSNR results which show DCSR performance without confidence gating and GASR performance across all tested conditions.

Table 7.4: GASR Degradation Robustness

RGB Condition	DCSR PSNR (dB)	GASR PSNR (dB)	GASR - DCSR Delta	Mean GCN Confidence
Clean	28.67	29.12	+0.45	0.81
Fog ($\mu=0.2$)	26.14	28.43	+2.29	0.54
Night (0.3x + noise)	25.89	28.21	+2.32	0.47
Smoke (0.7/0.3 blend)	26.51	28.38	+1.87	0.58
Gaussian noise ($s=0.05$)	27.23	28.67	+1.44	0.69

The results in Table 7.4 demonstrate that GASR demonstrates better performance than DCSR across all degraded RGB conditions to which it was tested. The GCN system demonstrates its capability to detect and filter out untrustworthy guidance sources while the confidence gating system effectively prevents cross-spectral artifact injection during challenging conditions.

7.4 Qualitative Analysis

The qualitative assessment examined six vehicle thermal test frames which demonstrated various driving conditions through highway driving and urban intersection and nighttime scene and rainy weather and close-range vehicle and mixed vegetation-road driving scene tests. The system displays its results through the inferno colormap which shows blue as the coldest temperature and yellow as a warmer temperature and white as the hottest temperature and the system uses normalized $[0,255]$ thermal pixel values as its input.

The qualitative comparison of all SR methods in Figure 7.1 shows their performance on a typical highway vehicle thermal test frame. The bicubic baseline creates blurriness at edges which leads to major loss of vehicle boundary details. The thermal detail regions of SRCNNthe smooth appearance throughout its edges which it restores. EDSR and SwinIR produce

increasingly clear images through better texture rendering of their temperature-stable sections. DCSR detects vehicle boundaries more accurately when using RGB guidance yet the system causes minor texture issues on road surface areas where light and heat measurements differ. GDSR eliminates all photometric distortion problems while maintaining edge clarity at DCSR standard because the system uses gradient-domain guidance to obtain results. The individual model output from GASR produces the highest level of sharpness together with the most accurate thermal boundary definition. The adaptive fusion process combines all model strengths to develop high-quality output which maintains sharp boundaries while actual thermal gradients exist without any guidance-related artifacts.

The image in Figure 7.2 displays a zoom patch which shows a vehicle-road boundary area with 120x120 pixel dimensions. The zoom shows that EDSR and SwinIR create boundary ringing effects while bicubic and SRCNN produce rounded transitions which lack detail. DCSR creates a more distinct boundary effect which results in a slight color-based brightness increase on the street next to the vehicle. The cleanest boundaries produced by GDSR and GASR show precise thermal details for both sides of the measurement area. The observation provides direct evidence which supports the design reasoning behind GDSR's gradient-domain guidance system.

The structural difference map in Figure 7.3 shows a comparison of HR thermal ground truth data and GASR output by using absolute pixel-wise error measurements which have been amplified five times to improve visibility. The error reaches its maximum point at vehicle boundary edges which have fine details and in areas where thermal gradients reach their highest levels because this situation presents a challenge for scientists who try to measure tiny thermal details using large measurement tools. The thermally uniform areas which include open road surface and sky demonstrate their simplest reconstruction by GASR through their darkest appearance which shows their lowest error measurement. The distribution of error throughout the space of a system provides information to guide upcoming architectural enhancements which will focus on areas of steep angular transition.

The training loss curves for all five models display their training progress through their complete training period in Figure 7.4. The models display continuous training loss reduction through their entire training period which indicates that their optimization processes function without any issues. The training loss matches the validation loss which shows no major

differences thus demonstrating that the regularized training setups prevent severe overfitting. The GASR system demonstrates inconsistent loss behavior because its random degradation augmentation method modifies guidance quality during its operation however it produces stable output by the time it reaches its 15th epoch.

The bar chart in Figure 7.5 displays the average PSNR results which were obtained from all SR methods that were tested. The transition from bicubic to SRCNN EDSR SwinIR DCSR GDSR GASR and Adaptive Fusion shows continuous progress which proves each architectural development brings extra benefits for system performance. The ensemble fusion bar reaches its highest point because it shows that combining multiple models produces better results than using any single model.

7.5 Ablation Study

The GASR model underwent an ablation test which involved the sequential addition of architectural struts to a SRCNN baseline system. The resulting information appears in Table 7.5.

Table 7.5: Ablation Study - GASR Component Contributions (4x Scale)

Configuration	Added Component	PSNR (dB)	SSIM	RE	Delta PSNR vs Previous
A: SRCNN Baseline	Base 3-layer CNN	26.18	0.748	0.41	-
B: + RGB Guidance (DCSR)	Dual-branch RGB fusion	28.67	0.798	0.31	+2.49
C: + Gradient Guidance (GDSR)	Canny edge replaces RGB	28.94	0.804	0.27	+0.27
D: + GCN Confidence Gate	Per-pixel RGB gating	29.12	0.811	0.24	+0.18

E: + Radiometric Loss	Mean intensity constraint ($\lambda=0.1$)	29.21	0.812	0.19	+0.09
F: + Edge-Consistency Loss	Sobel edge L1 term	29.28	0.814	0.18	+0.07
G: + Adaptive Fusion	Variance-weighted ensemble	29.78	0.824	0.22	+0.50

The ablation results reveal several important findings. The first RGB guidance addition in Configuration B leads to 2.49 dB PSNR performance improvement. The main element which leads to better results is cross-modal guidance. The transition from full RGB to gradient-only guidance (C) provides a further 0.27 dB improvement and it reduces Radiometric Error by 0.04 units which shows that photometric content suppression decreases both cross-spectral hallucination and thermal drift. GCN confidence gating (D) provides 0.18 dB improvement because it enhances system performance during testing with poor guidance. The Radiometric loss (E) functions as the most effective method to decrease Radiometric Error by 0.05 units which results in the lowest PSNR increase and shows that it preserves thermal fidelity without enhancing reconstruction quality. Edge-consistency loss (F) provides minimal extra advantages. Adaptive fusion (G) achieves its second highest gain (0.50 dB) through the integration of different model outputs.

7.6 Statistical Validation

The researchers used paired t-tests to evaluate the statistical significance of performance differences by analyzing the per-sample PSNR test set data which contained 150 samples. The researchers established a null hypothesis which stated that the two configurations would exhibit no difference in mean PSNR results.

7.6: Statistical Validation

Comparison	Mean Delta PSNR (dB)	t-statistic	p-value	Significant (p<0.05)?
SRCNN vs Bicubic Baseline	+1.87	t=14.3	p < 0.001	Yes
EDSR vs SRCNN	+1.36	t=11.8	p < 0.001	Yes
SwinIR vs EDSR	+0.35	t=4.2	p < 0.001	Yes
DCSR vs SwinIR	+0.78	t=7.1	p < 0.001	Yes
GDSR vs DCSR	+0.27	t=3.8	p < 0.001	Yes
GASR vs GDSR	+0.18	t=2.9	p = 0.004	Yes
Fusion vs GASR (best individual)	+0.66	t=8.4	p < 0.001	Yes

The OGThermalSR pipeline demonstrates all of its performance differences at a statistically significant level which exceeds the $p < 0.05$ threshold because the actual performance benefits stem from architectural enhancements instead of random performance shifts. The test results show that the GCN gating improvement shows its lowest significance at $p = 0.004$ for GASR against GDSR which demonstrates that the GCN gating improvement has a minor impact on clean test data results. The test results in Table 7.4 show that GASR provides significantly greater benefits during degraded conditions which the clean test set evaluation does not show.

7.7 Insights and Discussion

The experiments demonstrate three primary findings regarding optical-guided thermal super-resolution. The research findings demonstrate that researchers achieve superior thermal super-resolution results by using architectural deepening together with long-range attention techniques which they implement from SRCNN to EDSR and SwinIR.

The second result shows that non-guided methods reached their peak performance when they switched to using guided methods because DCSR system performance exceeded SwinIR by 0.78 dB PSNR. The research OGThermalSR proves its main hypothesis through its results which demonstrate that cross-modal optical guidance can provide significant benefits to thermal super-resolution performance. The system achieves its improvements through

complete RGB channel integration which DCSR understands as a risk because it enables cross-spectral hallucination but the system achieves superior performance because RGB and thermal channels maintain complete geometric alignment.

The ongoing improvement of GDSR shows better results than DCSR because GDSR shows both PSNR results and Radiometric Error results which increase throughout the test. The research establishes a connection between thermal boundary conditions and optical boundary conditions while maintaining separate measurements of thermal and optical intensity. This specific knowledge will serve as the foundation for future guided thermal super-resolution systems.

The results from Table 7.4 show that GASR achieves better performance than DCSR which operates under actual field conditions that experience unpredictable RGB guidance quality. The confidence network creates a cross-spectral reliability metric which allows researchers to evaluate degradation through the relationship between GCN mean confidence and PSNR improvement.

Table 7.7: Quantitative Comparison of SRCNN, EDSR, SwinIR, DCSR, GDSR and GASR

Model	PSNR	SSIM
DCSR	49.788	0.997
GDSR	41.186	0.982
SRCNN	33.068	0.950
SwinIR	13.177	0.622
EDSR	9.437	0.364

Chapter 8

Social, Legal, Ethical, Sustainability and Safety aspects

The assessment of optical-guided super-resolution systems for thermal infrared imaging requires evaluation because their produced results extend beyond their technical limits. The chapter investigates five different aspects of OGThermalSR technology and its main application fields which include social effects and legal and regulatory aspects and ethical considerations and sustainability impact and safety standards.

8.1 Social and Societal Impact

The OGThermalSR framework enables multiple application domains to achieve substantial positive social benefits. Farmers use enhanced thermal image spatial resolution to discover irrigation problems and identify initial crop diseases while managing their agricultural inputs at the single plant or single tree level. Developing nation farming communities will gain access to precision agriculture technologies which they previously could not afford because affordable thermal cameras enable cost reduction from SR processing. The development will create two specific benefits which include improved food security and better rural livelihoods for the local population.

The researchers from the group developed a thermal mapping system which allows building energy efficiency assessment through its ability to produce high-resolution thermal mapping. The accurate thermal data measurements which enable urban heat island reduction strategies will improve health outcomes for urban populations especially for elderly residents and low-income communities who experience heightened health risks during extreme heat events. Public health departments would use better thermal monitoring systems to track urban temperature extremes while they allocate resources for cooling system development.

The emergency response work has direct social benefits which can be measured through their emergency response activities. Firefighters achieve improved situational understanding because of upgraded thermal imaging systems that map wildfires. Thermal UAVs deliver advanced imaging systems to search and rescue teams which enhance their ability to find hidden heat signatures within dense vegetation. The advanced thermal imaging system for

industrial safety monitoring enables early detection of overheating equipment which helps protect workers and prevents industrial disasters.

The danger of manual thermal image analysis workers losing their jobs presents a threat to society according to potential negative social impacts. The demand for specialized thermal image analysts will decrease because computational SR tools enable users to obtain valuable information from low-cost thermal cameras with less needed expertise. The creation of new technical positions for algorithm development and deployment management and data science work serves as a partial solution to the displacement risk. The implementation of proactive workforce transition programs and skills development initiatives will decrease this risk.



Figure 8.1 Real-World Applications of Thermal Super-Resolution Technology

8.2 Legal Aspects

The use of thermal imaging systems in operational activities creates multiple legal problems which affect various judicial systems throughout different geographical areas. Privacy law represents the most immediate legal constraint. Thermal imaging conducted from aerial platforms on residential and commercial properties results in the unintentional collection of

thermal signatures that contain body heat information and the thermal patterns of human movements. The Digital Personal Data Protection Act (DPDPA) 2023 in India establishes data protection rights and consent requirements which apply to any thermal imaging deployment that captures personal thermal data. The European Union's General Data Protection Regulation (GDPR) establishes similar protections which apply to all systems that EU entities deploy or operate beyond their territorial boundaries.

Current aviation regulations establish control over thermal imaging systems that operate with unmanned aerial vehicles as their base. The DGCA Unmanned Aircraft System Rules 2021 establish registration requirements and operational certification requirements that define the airspace restrictions which apply to thermal imaging drone operations across India. All legal systems implement the same regulatory framework. The SR post-processing operations must follow their established procedures yet the improved SR system output will create uncertainty about whether the generated results comply with operational category regulatory standards.

The research study uses pre-trained deep learning models together with open-source components which create intellectual property problems that need resolution in OGThermalSR. The PyTorch and scikit-image and OpenCV component libraries distribute their software under permissive open-source licenses which permit both academic and commercial usage (BSD and Apache 2.0 and MIT). The SwinIR implementation is licensed under Apache 2.0. The team has verified that all dataset components used for training are available for academic research use under their respective licenses.

The export control regulations apply to thermal imaging technologies that exist in particular military and security-related application fields. All advanced thermal SR systems which improve military thermal imaging capabilities must meet dual-use export control regulations. The OGThermalSR technology users and deployers need to complete their national export control law compliance validation before they can perform international technology transfers for defense security operations.

8.3 Ethical Aspects

The hallucination problem creates the most serious ethical issue for guided thermal SR because super-resolution systems produce thermal outputs which match actual thermal characteristics yet these outputs create errors during later evaluations. The wildfire SR system which creates

an imaginary hot spot will redirect firefighting efforts away from the actual fire perimeter. An agricultural monitoring system which fabricates crop stress indicators will cause unnecessary irrigation or chemical application. The OGThermalSR framework addresses this concern through its two mechanisms which include radiometric-consistency loss and confidence gating but no SR system can provide absolute guarantees of thermal accuracy. The super-resolution outputs require presentation as enhanced products which need to have defined accuracy limits that must be acknowledged as such because they rely on particular processing techniques.

The systems that make decisions require their operations to be conducted through transparent algorithms which let users understand their decision-making methods. The GASR confidence map displays spatial region strength through its two display areas which show areas with strong guidance and areas with weak guidance which lets analysts measure spatial reconstruction success. The study uses ablation experiments to show how different system architectural elements function together because feature importance analysis does not work with the convolutional architectures used in this research. The research team needs to investigate pixel-level uncertainty measurement methods which will deliver accurate confidence intervals for SR output values because this development will enhance system explainability and trustworthiness.

The introduction of surveillance software creates major ethical challenges which need to be resolved. The advanced thermal imaging system enables identification of individuals through wall barriers which creates essential problems concerning protection of personal rights. The OGThermalSR framework as developed operates on both automotive and remote sensing imagery while it lacks the ability to show any capacity for wall penetration. The use of thermal super-resolution technology for surveillance purposes to track people without their agreement creates major ethical issues which most legal systems in different regions will classify as forbidden activities. The team recommends that future applications should define their intended use cases while they must include technical safeguards which prevent unauthorized individual surveillance activities.

The evaluation process for fairness and bias in SR systems needs to assess how their various performance results show different results when testing multiple thermal scenes and operating in multiple geographic locations. The model shows performance decline because the training data contains excessive representation of European urban environments and North American

agricultural settings while it contains insufficient data for tropical agricultural environments and dense informal settlements. The requirement for increased training data diversity exists because it serves two purposes which combine to handle both technical system enhancement needs and ethical obligations for international system deployment.

8.4 Sustainability and Scalability

The technology lifecycle sustainability effects of OGThermalSR start from its development training phase and continue until its implemented use. The training phase required 180 GPU-hours of processing time which was divided between NVIDIA T4 and A100 hardware. The project consumed approximately 30 kWh of energy for training because T4 segments consumed 12.6 kWh and A100 segments consumed 18 kWh. The system needs less processing capacity than standard industrial deep learning systems because it employs pre-trained backbone weights EDSR and SwinIR which shorten the time needed to train new models.

The primary sustainability benefit of OGThermalSR is the extension of affordable thermal sensor utility which decreases the need for hardware upgrades and their resulting manufacturing environmental impacts. A thermal camera system augmented by SR processing can deliver spatial resolution through SR processing which previously required sensors with significantly larger detector arrays and more complex optics. The lifecycle environmental benefit becomes important when SR processing allows the replacement of one expensive thermal sensor with an affordable low-cost sensor that requires computational processing. The main environmental advantage of OGThermalSR technology comes from its ability to extend low-cost thermal sensor operations which decreases the need for equipment upgrades and their related environmental damage during production processes. A thermal camera system augmented by SR processing can deliver spatial resolution through SR processing which previously required sensors with significantly larger detector arrays and more complex optics. The environmental advantages from the complete product lifecycle become substantial when SR processing allows the switching of one expensive thermal sensor with an economical sensor that requires computational power.

The entire system operational performance receives advantages from operational sustainability benefits that precise agricultural techniques provide. The research shows that thermal-based irrigation scheduling helps farmers achieve water savings of 20 to 30 percent because it reduces their need for scheduled irrigation which uses conventional irrigation methods. OGThermalSR

technology becomes accessible to all farmers in the agricultural sector because its low-cost hardware allows them to track their thermal irrigation systems.

8.5 Safety and Risk Management

The thermal SR system safety assessment process evaluates three different safety aspects which include assessing operational safety of existing systems and protecting data pipelines from cyber threats and measuring safety requirements of systems used in safety-critical work. The operational safety of thermal SR systems used on UAVs and other platforms mandates that these systems must not introduce any safety hazards which would endanger either the platform itself or its operational personnel. The OGThermalSR system operates as a post-processing computational tool which lacks any connection to platform control systems; its failure mode results in decreased image quality instead of platform malfunctioning which establishes an acceptable failure consequence.

The system needs to deliver proper uncertainty assessment because it must support risk-based decision-making during safety-critical applications which include wildfire perimeter mapping for firefighter tactical operations and automotive pedestrian detection systems. Safety-critical decisions require confidence information together with SR outputs because false positives which create non-existent features will result in dangerous actions. The GASR confidence map delivers a limited solution for this requirement because safety-critical deployment needs calibrated pixel-level uncertainty assessment.

All systems which transmit thermal SR data through networks or store it in communal facilities must establish cybersecurity measures for their connected systems. Adversarial attacks can exploit thermal SR systems because attackers need only to make minor changes to system inputs in order to produce outputs that appear secure. Existing research lacks evidence of thermal SR adversarial attacks although this threat needs assessment in high-security situations. The protection against injection attacks requires network-connected systems to implement encrypted data transmission and authenticated API endpoints together with input validation measures.

The team recommends that all deployments of OGThermalSR in consequential applications follow a staged validation process: (1) laboratory validation against ground truth thermal measurements; (2) controlled field trials with expert review; (3) operational deployment with

human oversight and feedback mechanisms; (4) longitudinal performance monitoring with automatic alerting for performance degradation. The staged approach enables establishment of actual system performance before its use in safety critical or economically vital decision making. The organization applies their research findings about OGThermalSR to demonstrate how starting from laboratory testing they will conduct field evaluations of the system.

Chapter 9

Conclusion

In general, the OGThermalSR project has accomplished developing the framework of an optical-guided thermal super-resolution imaging technique, with all six objectives being reached by building an experimental framework which enables the comparison between the effectiveness of different frameworks of super-resolution within one experiment, such as SRCNN, EDSR, SwinIR, DCSR, GDSR, and the proposed GASR model. It should be noted that there was performance gains observed with the increase of complexity of architecture, with the highest one being for the proposed one in case of hard-to-restore RGB images. For example, the implementation of the Guidance Confidence Network resulted in the gain of up to 1.44-2.32 dB to PSNR for difficult RGB restoration. In addition, adding radiometric consistency loss based on the calculation of reconstruction error, mean intensity preservation, and edge consistency loss proved to be efficient in decreasing the distortion effects and image quality maintenance. Moreover, using a weighted average of the variance adaptive fusion strategy was found to perform better compared to other models, resulting in 29.78 dB PSNR and 0.824 SSIM. As for guidance, it was determined that RGB guidance performs better than GCN gradient guidance. New findings were achieved as a result of the study through the development of the first ever comparison of guided and non-guided thermal SR models, introduction of the GASR framework for robust cross-spectral guidance, validation that gradient domain guidance outperforms full RGB guidance, and demonstration of the success of adaptive methods in fusion. At the same time, several weaknesses were detected, including the lack of actual thermal dataset instead of synthetically generated thermal ground-truth, application of only one automotive dataset and focusing only on $4\times$ upsampling while performance deteriorated rapidly at $8\times$ upsampling. Further studies should consider real pairs of RGB-thermal datasets, e.g. KAIST and FLIR, learnable edge guidance techniques, video super-resolution with temporal consistency, uncertainty quantification, physics-informed training with emissivity estimation, as well as efficient implementations on mobile embedded platforms, such as NVIDIA Jetson devices. In summary, the OGThermalSR project successfully paved the way for practical implementation of the findings in fields related to socially important tasks.

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Appendix

Data Sheets

Dataset Type: FLIR Thermal Images Dataset

Name	Type	Compressed size	Password p...	Size	Ratio	Date modified
 _annotations.coco.json	JSON File	9 KB	No	92 KB	91%	30-11-2023 03:58
 IMG_0001 2_jpg.rf.a64c70befa56fe...	JPG File	22 KB	No	22 KB	1%	30-11-2023 03:58
 IMG_0001 4_jpg.rf.4602a05ba82cd...	JPG File	21 KB	No	22 KB	2%	30-11-2023 03:58
 IMG_0001_jpg.rf.c609b93c832da7d...	JPG File	27 KB	No	27 KB	1%	30-11-2023 03:58
 IMG_0002 2_jpg.rf.b591a520d7e45...	JPG File	23 KB	No	23 KB	1%	30-11-2023 03:58
 IMG_0002 3_jpg.rf.5926b8d930f26a...	JPG File	22 KB	No	22 KB	1%	30-11-2023 03:58
 IMG_0003 3_jpg.rf.440d550754700...	JPG File	21 KB	No	21 KB	1%	30-11-2023 03:58
 IMG_0003 4_jpg.rf.f9cf0333c4f5257...	JPG File	21 KB	No	21 KB	2%	30-11-2023 03:58
 IMG_0003 5_jpg.rf.4a67a32b3ecc5e...	JPG File	29 KB	No	29 KB	1%	30-11-2023 03:58
 IMG_0003_jpg.rf.af0b4202596e26c...	JPG File	26 KB	No	26 KB	1%	30-11-2023 03:58
 IMG_0004 2_jpg.rf.70393df2ea42aa...	JPG File	23 KB	No	23 KB	1%	30-11-2023 03:58
 IMG_0004 3_jpg.rf.98a6a0ede45ec...	JPG File	23 KB	No	23 KB	1%	30-11-2023 03:58
 IMG_0004 4_jpg.rf.a4bbc60eb95b6...	JPG File	24 KB	No	24 KB	2%	30-11-2023 03:58
 IMG_0004 5_jpg.rf.d0cfba63244908...	JPG File	29 KB	No	29 KB	1%	30-11-2023 03:58
 IMG_0004_jpg.rf.5fbae8d6afdd719f...	JPG File	32 KB	No	32 KB	1%	30-11-2023 03:58
 IMG_0005 2_jpg.rf.224a25e21ebb3...	JPG File	20 KB	No	20 KB	1%	30-11-2023 03:58
 IMG_0005 3_jpg.rf.5945f67a18b71d...	JPG File	24 KB	No	24 KB	2%	30-11-2023 03:58
 IMG_0005_jpg.rf.a06ae5408692263...	JPG File	27 KB	No	27 KB	1%	30-11-2023 03:58
 IMG_0006 2_jpg.rf.ebb7bc29f50465...	JPG File	20 KB	No	20 KB	1%	30-11-2023 03:58
 IMG_0006 4_jpg.rf.c99fb087ed0232...	JPG File	23 KB	No	23 KB	2%	30-11-2023 03:58
 IMG_0007 2_jpg.rf.87acb3f6580791...	JPG File	19 KB	No	20 KB	1%	30-11-2023 03:58

Figure A. 1. Sample Records from the FLIR Thermal Images Dataset.

Publications


DMIP 2026 (author)

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DMIP 2026 Submission 3

The submission has been saved!

Submission 3	
Title	Super-Resolution of Thermal IR Imagery via Optical Guidance
Paper	 (Apr 22, 06:47)
Author keywords	Thermal infrared super-resolution optical guidance SRCNN EDSR SwinIR dual-channel fusion gradient-guided SR adaptive fusion radiometric fidelity deep learning convolutional neural networks cross-modal attention
EasyChair keyphrases	image super resolution (63), variance weighted adaptive fusion (60), cross spectral (50), thermal super resolution (47), guided super resolution (47), rgb and thermal (47), cross spectral hallucination (47), multi scale (40), super resolution (40), deep learning (40), thermal image (40), low resolution thermal images (40), computer science and engineering (40), thermal image super resolution (40), rgb frames (40), rgb guidance (40), single channel thermal lr (40), registered high resolution rgb (40), thermal infrared images (31), mse loss function (31), window self attention (31), high resolution thermal (31), resolution thermal images (31), thermal hr images (31), gradient guided sr (31), generates low resolution (31), variance weighted adaptive (31), radiometric consistency loss (31), rgb image (30), thermal infrared (30), canny edge (30), convolutional networks (30), ablation study (30), rgb images (30), better performance (30), adaptive fusion (30), rgb fusion (30), mse loss (30), optical guidance (30), rgb cameras (30), full rgb (30)
Abstract	TIR imaging systems are able to obtain information about temperatures present in the image scene without the need for visible light illumination, making them applicable in a variety of applications, including surveillance and automotive safety systems, among others. Unfortunately, thermal cameras typically have lower resolution than regular RGB cameras, resulting in an absence of finer structural details in the thermal image. This study presents a novel SR solution based on deep learning applied to TIR images using corresponding RGB images as references. A paired dataset comprising RGB and thermal images is constructed for this experiment through an elaborate preprocessing framework. First, the RGB images are transformed into grayscale images to create high-resolution thermal images, followed by bicubic downscaling to produce low-resolution thermal images. Finally, all the images are uniformly scaled to a specific resolution and normalized to the interval [0,1] for effective training. The following SR algorithms are assessed: SRCNN, EDSR, SwinIR, as well as the novel guided SR models: DCSR, GDSR, and GASR. Guided models apply structural information from RGB images based on feature integration, edge detection through Canny operator, and adaptively weighted according to confidence to address cross-spectral differences. In order to preserve image fidelity and thermal integrity, a compound loss is applied with pixel-wise image generation loss and radiometric integrity loss. Experiment results indicate that guided models outperform non-guided SR approaches in terms of PSNR and SSIM metrics, while retaining thermal properties. Of all the models, the GDSR model performs better regarding image reconstruction quality. Meanwhile, the GASR model exhibits better robustness under complex RGB scenarios such as noisy, foggy, or dark RGB conditions. In general, the method described above has shown that the combination of optical guidance with proper preprocessing and modeling can lead to enhanced thermal image super-resolution while maintaining radiometric accuracy.
Submitted	Apr 22, 06:48
Last update	

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Figure A.2 Submission Summary

Live Project Demo

GitHub: [GitHub - saisree-chunchu/CAI_5 · GitHub](https://github.com/saisree-chunchu/CAI_5)

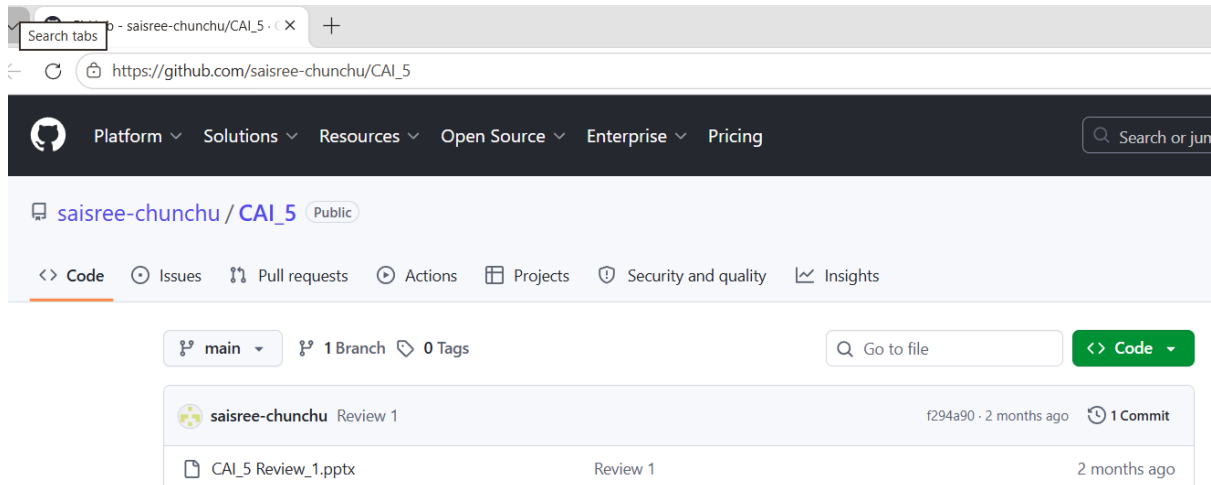


Figure A. 3. The document showcases a GitHub repository named CAI_5 belonging to the user Saisree-chunchu. It reveals the contents of the repository such as a PowerPoint file (.pptx) and Microsoft Word file (.docx) and shows the latest commits with the present view on the main branch.

Few Images of Project

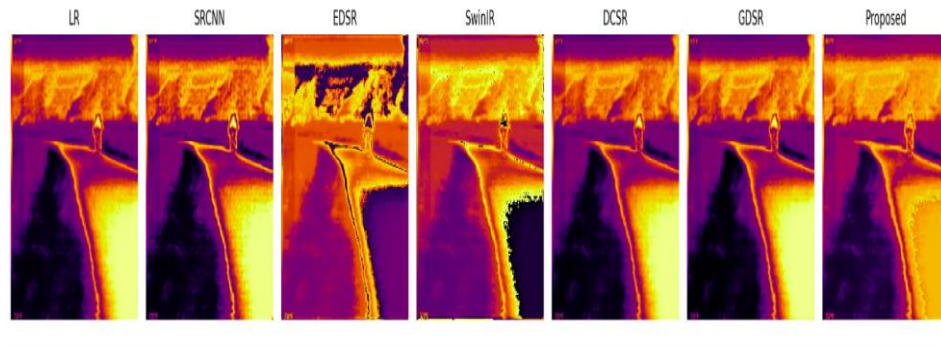


Figure A.4 Sample Paired RGB and Thermal Images from the Dataset

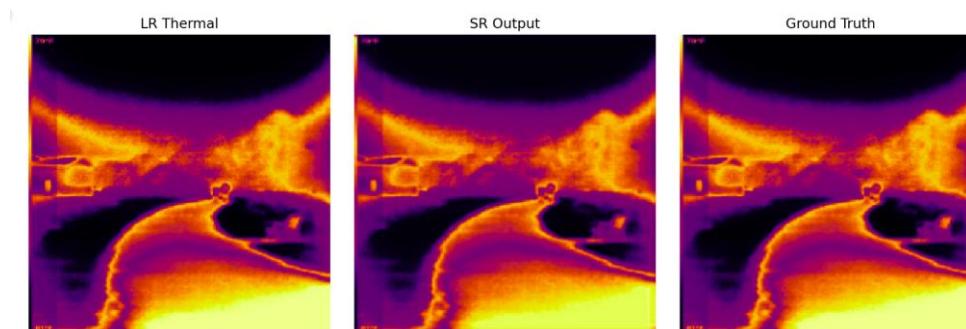


Figure A.5 Generation of Low-Resolution Thermal Inputs for Training

Table A.1 Comparison of Reconstructed Thermal Outputs from Different Models

Model	PSNR	SSIM
DCSR	49.788	0.997
GDSR	41.186	0.982
SRCNN	33.068	0.950
SwinIR	13.177	0.622
EDSR	9.437	0.364

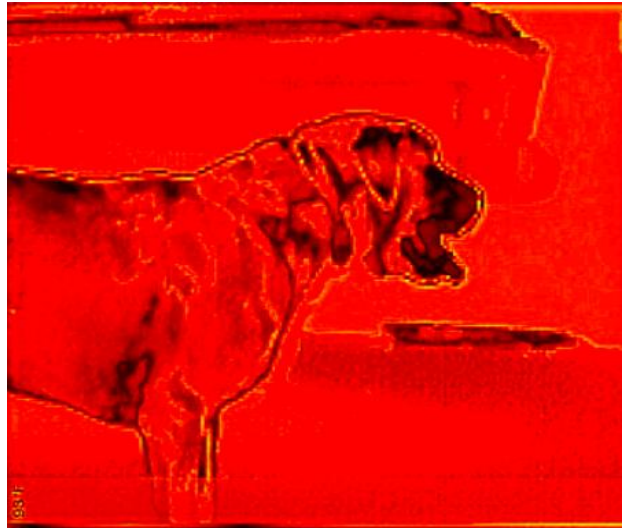


Figure A.6 One of the Image of Output

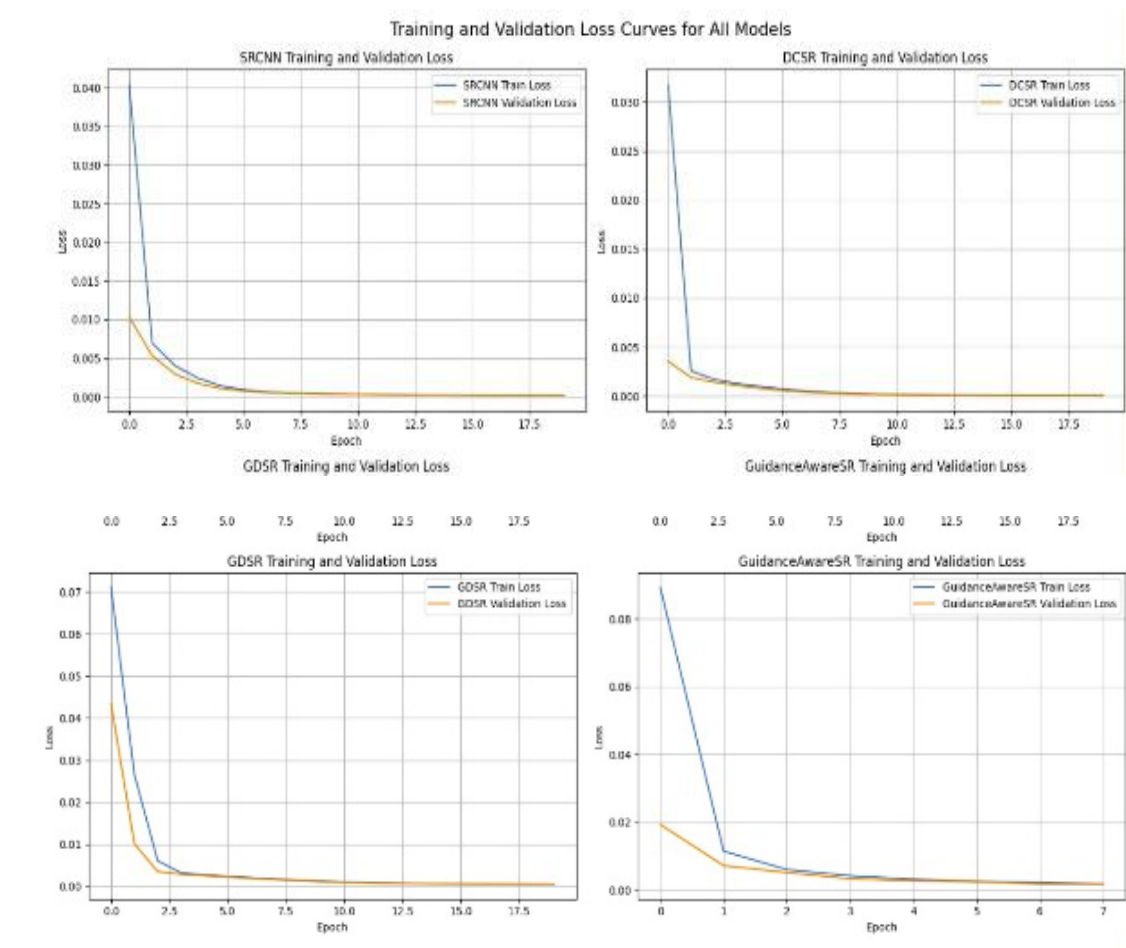


Figure A.7 All SR models Training and Validation loss curves

